The Component Companion

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Illustrated Electrical Symbols and

the Cognitive Roles They Map To

A bench guide to the component library of

CAS PCBA Engineer

Little Henderson

Noetic Compass Press

2026

The Component Companion
Illustrated Electrical Symbols and the Cognitive Roles They Map To

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Every parameter value in this guide is an engineering design choice. None of them is a measurement taken from a person, an AI model, or a biological system. Nothing here is a safety certification, a measurement instrument, or a claim about how any particular mind works.

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Every component in the bin, drawn as an engineer
would draw it,
and read as a mind would use it.

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Preface

The main volume teaches you to design cognitive circuits. This one is the reference you keep open beside it: the part bin, drawn.

It exists because of a specific problem. A schematic in a design chapter appears once, in the middle of an argument, doing one job. Later, at the bench, you need the same part again — and what you actually need is not the argument. You need the shape, the job, the mapping class, and the trap. That is four lines, and it should be four lines you can find in ten seconds.

So every entry in this guide is built the same way, and it is short on purpose:

- **The symbol**, drawn to the same conventions the main volume uses.
- **One electrical sentence** — what the part does in a circuit.
- **One cognitive sentence** — the role it plays in a cognitive circuit.
- **One worked example**, with real CAS state literals.
- **One guardrail** — the specific way this mapping goes wrong if you push it.

One badge per entry, and there are only three. Every mapping carries exactly one label, and the label answers one question: how much of the electrical behaviour actually transfers?

- **ISO — exact structural isomorphism.** The functional roles correspond one-for-one and the governing relation has the same form. The mapping is structural, not decorative.

- **ENG — engineered analog.** The role match is explicit, but you must stipulate a cognitive quantity, control input, or threshold before it means anything. The number is a design choice, not a discovery.
- **PED — pedagogical analogy.** The resemblance teaches, and that is all. Nothing quantitative transfers.

A fourth label would be a way of avoiding the question. If a mapping is *almost* ISO, it is ENG, and the badge says so.

What this guide is not. It is not a claim that minds are circuits, and it is not a claim that these mappings have been validated against anything. It is a design vocabulary with the boundaries drawn in. Every plate tells you where its own analogy stops. That is the whole point of printing the guardrail line.

How to Use This Guide

Three ways in, depending on what you are doing.

At the bench, mid-build. Go straight to the chapter for the part in front of you — Chapters 1 through 7 cover the symbol plates, one family per chapter. Read the symbol, the role, the example, the guardrail. Four lines.

Studying the framework. Read the chapters in order. Chapter 1 gives you the legend everything else assumes: ports, typed nets, forward and feedback arrows, the reference node, and the Domain colour key. The plates build on it. Chapter 10 compresses all forty-two entries into a single table you can memorise, and that table is the real deliverable — the plates are how you get there.

Reading a schematic. Chapters 8 and 9 take the ten worked cognitive schematics — from the single-transistor atom up to the power-management board — and show what each wire is carrying. Those are the same circuits the main volume designs, redrawn at reference size.

Printing the cards. The companion deck reproduces the whole guide on fifty-six cut-out cards, seven sheets of US Letter. Chapter 11 tells you how to print them so the text stays the size it was designed at.

The maintenance notes — the frozen inventory, the notation rules, and the open review items — live in Appendix A of this guide. If you are extending the library rather than reading it, start there.

How to Read These Plates

A circuit symbol is a compact statement about a **job**. The electrical part and the cognitive part need not share a physical substrate. The mapping is one-to-one only when their **functional roles** and governing relation line up.

Each entry therefore carries one of three labels:

- **ISO — exact structural isomorphism.** The functional roles correspond one-for-one, and the relation has the same form.
- **ENG — engineered analog.** The role match is explicit, but the design must stipulate a cognitive quantity, control input, or conservation rule.
- **PED — pedagogical analogy.** The resemblance helps teach. No quantitative relation transfers.

These labels apply to the **mapping**, not to the existence of the electrical component. All parameter values shown in this guide are design choices. They are not measurements from a person, an AI model, or a biological system.

The six reading rules

1. **Read the shape first.** It tells you what the part does.
2. **Read the wire label second.** A cognitive wire carries a CAS state or an explicitly named control quantity.
3. **Solid means forward. Dashed means feedback.** Arrow direction describes signal flow, not attraction or preference.

4. **Color reinforces Domain.** Shape and text carry the meaning, so every plate survives grayscale printing.
5. **Separate the four CAS axes.** Noetic says *what*. Domain says *where*. Drive says *why*. Register says *how*.
6. **Do not force one part onto one Noetic.** Components implement roles. One role may act on Noetic, Domain, Drive, Register, or a board-level control quantity.

Anti-drift checks

- `^3` **Negative** is repulsion. `^6` **Male** is decomposition. An audit or breakdown of an argument is `^6`, not `^3`.
- `^8` **Above/Cause** is an Idea that triggers a different Idea to come to Mind. It is not deductive reasoning. The arriving Idea is `^9` **Below/Effect**.
- + means co-presence. $\times$ requires a named emergent result. $|\langle\ldots\rangle$ specifies register occupancy without claiming integration.

Part I

Reading a Human Being

Four kinds of experience, two languages, one legend

Chapter 1

The Four Kinds of Human Experience

Chapter 1.

Before any symbol: what this book is actually about.

You already describe your own life in four different ways, and you switch between them without noticing. Ask someone how their week was and watch which answer comes out.

> "I finally worked out why the deal fell through." — **mental**
>
> "I've been miserable since Tuesday." — **emotional**
>
> "My shoulders have been up around my ears." — **physical**
>
> "I don't know what any of it is *for* any more." — **spiritual**

Same week. Four true answers. They are not four ways of saying one thing, and they are not interchangeable — a person can be intellectually clear, emotionally wrecked, physically fine, and utterly without purpose, all at once.

This book is organised around those four. They have a precise place in the framework behind it, and that is the first thing to see.

1.1 The four are an axis, not a list

The Cognitive Architecture System describes any mental event on four axes at once:

Axis	The question it answers
Noetic	*what* is happening
Domain	*where* — principled, cognitive, evaluative, or operational
Drive	*why* — which motive is powering it
Register	*how* — in what manner it is experienced

Register is the fourth. It has exactly four values, and they are the four answers above:

Register	Human word	What it is like from the inside
`|A` **Analytical**	**mental**	words, reasons, sequences, arguments you could write down
`|f` **Affective**	**emotional**	felt valence — drawn toward, pushed away, heavy, light
`|E` **Embodied**	**physical**	sensation, posture, breath, gut, the body's own knowing
`|T` **Transcendent**	**spiritual**	meaning, purpose, what it is all for; the sense of being oriented

That is the 1:1 this book is built on. Mental is `|A`. Emotional is `|f`. Physical is `|E`. Spiritual is `|T`. Not a metaphor, not an analogy, not a decorative parallel — the framework's own axis, read in ordinary human language.

Two consequences follow immediately, and both matter.

First: a single event is usually present in several registers at once, and the registers disagree. That is not confusion; it is the nor-

mal condition. Someone can know perfectly well, in |A, that a relationship is over, and still reach for their phone in |E every evening at six. The analysis finished; the body had not been told. Naming the registers turns a puzzling contradiction into a description: *this is resolved analytically and unresolved in the body.*

Second: a register you cannot name is a register you cannot work on. Most people who say "I'm fine" are reporting |A and not noticing |E. Most people who say "I can't think straight" are flooded in |f and blaming |A. The four words give you somewhere to point.

1.2 The same hour, in four registers

Here is one ordinary hour, written four times. Nothing about the events changes.

A woman is clearing out her late father's flat. She finds his coat.

|A — mental. *"The tenancy ends on the 30th. I need to decide about the furniture, and I should call the council about the meter reading."* Plans, sequences, administrative facts. The mind doing what a mind does when it wants a job it can finish.

|f — emotional. *"I miss him so much it makes me angry."* Valence. Not a thought about missing him — the thing itself, heavy and warm and unbearable in turns.

|E — physical. *"My throat closed when I opened the wardrobe. I've been standing in the same spot for ten minutes."* Sensation and posture, arriving before any of it has been put into words. This register often knows first.

|T — spiritual. *"He kept this coat for forty years. What was he holding on to? What am I?"* Meaning and orientation. Not a feeling and not a plan: the question of what a life is for, asked from inside one.

The registers are not stages and not a hierarchy. They are concurrent. A book that only ever says "she felt sad" has told you one quarter of that hour, and it has left out the quarter that usually de-

cides what happens next — the body, which was already standing still in front of the wardrobe.

1.3 Why a component library needs this

Because the components behave differently depending on which register they are working in.

A capacitor is retention. In `|A` it is a memorised fact you can still recall. In `|f` it is a grudge that has not discharged. In `|E` it is a flinch your hands do before you know why. In `|T` it is a vow. Same component, same job — retain what arrived — and four completely different human situations.

A resistor limits flow. In `|A` it is counting to ten. In `|f` it is refusing to let someone see how much they hurt you. In `|E` it is the breath you slow on purpose. In `|T` it is a discipline you keep because of who you intend to be.

So every entry in this book is written the same way, and it starts where you live:

- **The experience** — a real situation, told plainly.
- **What is happening** — the same situation read in CAS terms.
- **The part that matches** — the electrical component, and why the correspondence holds rather than merely rhymes.
- **Where it breaks** — the line past which the analogy stops being true.

And where a mapping genuinely holds across all four registers, all four are given, because that is the strongest evidence a mapping is structural rather than decorative.

1.4 How strong a claim each mapping makes

Not every pairing is equally deep, and pretending otherwise would make the whole book useless. Each carries exactly one badge.

- **ISO — exact structural isomorphism.** The roles correspond one-for-one and the relation has the same form. These are the mappings that survive being pushed hard.
- **ENG — engineered analog.** The roles match, but only once you stipulate a quantity, a threshold, or a control input. The number is a design choice you make, not a fact you discover. Most mappings are this, and saying so is what keeps the ISO ones credible.
- **PED — pedagogical analogy.** The resemblance teaches and stops there. Nothing quantitative transfers.

One badge per entry. Never a hybrid. If a mapping is *almost* ISO, it is ENG, and the badge says so.

1.5 The one-sentence version

Your mental life is `|A`, your emotional life is `|f`, your physical life is `|E`, and your spiritual life is `|T` — and a circuit that handles one of them can be read as handling that whole quarter of a person.

Everything that follows is that sentence, made specific forty-two times, and then walked through eight real situations from beginning to end.

Chapter 2

The Notation and the Symbols Enforce Each Other

Chapter 2.

There are two ways to write down a mind in this framework, and they are not two subjects. They are two views of one structure, and each one's rules hold the other one honest.

The notation writes an experience as an expression.

```
^8^C_{F2}|A  +  ^5^P_{F7}|T   ->   ^9^C_{F2}|A
```

A trigger, in the cognitive domain, powered by the Wisdom drive, in the analytical register. Meeting accumulated composition in the principled domain, powered by Purpose, in the transcendent register. Landing as an arrival, cognitive, Wisdom, analytical.

The symbols draw the same thing as a circuit. The `^8` is the Cause terminal. The `^5` is the Bias terminal. The `^9` is the Effect. The whole event is one transistor, and the wire labels carry the same clause the expression carries.

One is algebra. The other is topology. Anyone who has learned both finds that each one catches mistakes the other cannot see — and that is not a coincidence. It is the point.

2.1 What the notation enforces on the drawing

An expression that is illegal cannot be drawn, and this is where the notation does its work.

Admissibility. Every operator has a defined set of legal source-and-target pairs. The identity operator admits only the diagonal. This is a rule about *which crossings may occur*, and on a schematic it appears as a diode: a one-way gate that can be drawn in exactly two orientations, one of which is legal. If your drawing shows a crossing the notation does not admit, the drawing is wrong — not unconventional, wrong.

Involution is a fixed set of pairs. `^2`↔`^3`, `^5`↔`^6`, `^8`↔`^9`, and the self-duals `^0`, `^1`, `^4`, `^7` which are unchanged. This one rule kills more bad diagrams than any other. It is why an op-amp's inverting input is arithmetic subtraction and **not** the involution of a state; why a bride rectifier can never turn `^3` into `^2`; and why "the opposite of doubt" is not a component, because `^1` Mind is self-dual and has no opposite to convert to.

A gate needs a threshold. Continuous states live on the open simplex; there is no true and false in them until somebody says where the line is. So Boolean gates in a diagram must be preceded by an explicit thresholding stage. A drawing that feeds raw activation into an AND gate has drawn something that cannot exist.

Registers are specified, not assumed. $+$ is co-presence. $\times$ needs a named emergent result. $|\langle\ldots\rangle$ says a state is present in these registers and claims nothing about how they interact. Three different drawings, and the notation will not let you blur them.

Weights and reachability are arithmetic. A recipe's theoretical yield is a minimum over ratios; a fuse threshold has to sit below every upstream ceiling or it can never fire. These are not stylistic preferences. They are inequalities, and a drawing that violates them is a board that tests vacuously safe.

2.2 What the drawing enforces on the notation

The other direction is just as real, and it is the one that catches overclaiming in prose.

A wire carries one signal. If you cannot name what a net carries, you have not specified the circuit — and by the same rule, if your sentence contains three feelings bundled into one word, you have not specified the experience. "I was a mess" fails the net-label rule. Not morally; *structurally*. It cannot be traced, and therefore nothing downstream can be reasoned about.

Terminals have fixed identities. A component's input is its input. This is why the Cause/Bias/Effect labelling on the transistor is a genuine constraint rather than a convenience: bias shapes the transition, and anything that reaches back and moves the bias is a different component, drawn as feedback. The same rule outlaws a very common human sentence — *"he made me feel X"* — because it routes the large path back into the small control line, and the notation has no component for that.

Nothing is created at a wire. A schematic cannot manufacture a quantity between two components. So a mapping that claims a conversation *produced* a feeling out of nothing has drawn something impossible. It may have *triggered* one — that is `^8` meeting `^5` and landing `^9`, which is a legitimate and completely different claim.

Every path has a return or it does not. Solid is forward, dashed is feedback, and drawing the second as though it were the first is the single most common way people misread their own behaviour: the returning signal arrives looking like a fresh cause.

Protection must be reachable. Already stated, and worth stating twice, because it is the rule that converts a diagram from reassuring to true.

2.3 They are not translations of each other

The temptation is to read the two as the same sentence in two fonts. They are not, and the difference is worth stating carefully, because it is where most of the value lives.

They overlap. For a great many events you can write it either way and the two agree completely — a trigger meeting a substrate and landing an effect is one expression and one transistor, and neither says anything the other leaves out. That agreement is what makes the pair usable at all.

They also diverge, in both directions. Some things the notation states precisely, the drawing can only gesture at:

- **Admissibility relations.** The notation carries a defined set of legal source-and-target pairs per operator. A schematic shows *a* diode in *an* orientation. It does not show you the whole relation — only the one crossing you drew.
- **Involution structure.** `^2`↔`^3`, `^5`↔`^6`, `^8`↔`^9`, with `^0`, `^1`, `^4`, `^7` self-dual. That is a small, closed, checkable fact. In a drawing it is scattered across whatever components happen to be on the page.
- **The open simplex.** Every state is a non-negative vector summing to one. The notation can say that in a line. No schematic can say it at all — it is a constraint on the whole space, not a feature of any wire.
- **Register specification versus register co-presence versus integration.** `|⟨…⟩`, `+`, and `×` are three different claims about the same three terms. A drawing has no reliable way to distinguish them; the notation will not let you confuse them.
- **Recipes and yields.** Proportional prerequisite weighting, a limiting reagent, an efficiency between zero and one. This is arithmetic, and the drawing is not the right instrument for arithmetic.

And some things the drawing says better than the notation ever will:

- **Ordering and topology.** Whether a stage comes before or after another, what feeds back into what, which path is the main one. An expression is a statement; a schematic is a *shape*, and shapes show structure that a line of notation makes you reconstruct.
- **Isolation.** A relay's two sides are deliberately not connected. There is no elegant way to write "these two things affect each other without touching". You draw it.
- **Protection and position.** Where a fuse sits relative to a clamp — and therefore whether it can ever fire — is immediately visible on a schematic and easy to miss in an expression.
- **Multiplicity at a glance.** Six components sharing one bias node is one picture and a long sentence.
- **Reference and scale.** What everything is measured against, and where the zero is.

So: they enforce each other, and they are not interchangeable. Each holds expressions the other does not. When they agree you have a well-formed description. When only one of them can say the thing you mean, that is not a failure — it is information about *what kind of thing* you are describing. Ordering, isolation and constraint-space belong to one language; admissibility, involution and arithmetic belong to the other.

The practical rule for this book: **write the human situation in prose, the structure in whichever language carries it, and the check in the other one.** If you cannot draw it, ask whether the notation is even self-consistent. If you cannot write it, ask whether the drawing is missing a component that the notation would have forced you to name.

2.4 The schematic is a statement of intent

There is a third difference, and it is the one that matters most for anybody using this library on a real person or a real system.

The notation reports. The board commits.

An expression tells you what a state *is*. `^8^C_{F2}|A -> ^9^C_{F2}|A` says: a trigger met a substrate and an arrival landed. It is a description, and it is true or false about the world.

A schematic tells you what somebody **intends to happen**. Every symbol on it was placed by a designer; every parameter on it was chosen. `sat=0.8` is not a measurement — it is a decision about how high the output is allowed to go. `threshold=0.05` is not a discovery — it is a decision about where the boundary sits. β`=2.0` is not an observation — it is how much gain the architect decided this stage should have.

The notation has nowhere to put intent. The board is nothing but intent. That is the division of labour:

The notation says	The schematic says
what the state is	what the designer is trying to cause
this trigger landed this arrival	I have *biased* it this way, on purpose
`^3` repulsion and `^2` attraction are an involution pair	I have placed a guard here so the reverse crossing cannot occur
a fuse has a threshold	I have set it *here*, below the upstream clamp, so that it can actually fire
feedback exists	I have delayed it by τ and limited it to β, deliberately

So **CAS PCBA is how you write down how the notation is being used**, and what effect the architect intends. Read a schematic and you are reading a set of decisions. Read an expression and you are reading a claim about the world. The two can both be correct and still be about completely different things.

What intent looks like on the page

Every one of these is a decision, and none of them is in the expression:

- **Which parts are present at all.** An expression mentioning feedback does not tell you whether anyone built a return path.
- **Where they sit relative to each other.** A fuse upstream of a clamp and a fuse downstream of it are different designs with the same part list.
- **The numbers.** Thresholds, ceilings, gains, time constants, tolerances. All chosen, all stated, all design.
- **The default and the failure state.** What happens when the select line floats, when the input is unreadable, when the supply sags. A designer who has not stated this has not finished.
- **What is isolated from what.** A relay's two sides; a transformer's two windings. Isolation is an intention, and it is invisible in prose.
- **What is labelled.** A net label is a designer telling you what they believe that wire carries.

The three badges are a statement of how strong the intent's claim is

This is the cleanest reading of the badge system, and it is worth stating in these terms.

- **ISO — exact structural isomorphism.** The designer is claiming the correspondence is *structural*: the roles line up one-for-one and the relation has the same form. This is the strongest intent a mapping can carry.
- **ENG — engineered analog.** The designer is *stipulating*. The correspondence is meant, and the number is a decision they are making, not a fact they found. Most mappings are this, and saying so is what keeps the ISO ones honest.

- **PED — pedagogical analogy.** The designer is claiming *nothing* quantitative. The resemblance teaches and stops.

One badge per entry, never a hybrid, because "almost ISO" is not a third thing — it is ENG, and the badge has to say so. A hybrid badge is a designer declining to state the strength of their own claim.

Guardrails are the designer stating their own envelope

Every entry in Chapters 4–9 ends with a line beginning *Where it breaks.* That is not modesty and not hedging. It is **the operating envelope**, which is a standard engineering artifact: the range of conditions within which the designer warrants the behaviour, and the specific way it fails outside that range. A datasheet without an absolute-maximum section is not a more confident datasheet.

Which is why the guardrail is the most useful line on the page. It tells you where the designer's intent stops being valid — and therefore where *your* design has to take over.

Whose board are you running?

The reason this distinction is not academic.

A person's behaviour can be read as a board — rails, thresholds, fuses, references, return paths. The next question is the one that does the work: **who designed it?**

Some of it you designed. Somebody who has decided, deliberately, how much sleep they need, what they will not do on the phone, when they leave the room — that person has authored a board, and it does what they intended.

Much of it you did not. A rule inherited from a parent is a component somebody else placed. A reference set in a house where the atmosphere had to be monitored is a zero point you did not choose. A fuse whose trigger condition was named by somebody else will not fire when you need it. You can be running a board designed by

a person who is not you, for a purpose that was not yours — and the board runs perfectly. That is what makes it hard to notice.

Reading your own behaviour as a schematic is how you find out which parts you chose. Not to reject the inherited ones — most of them are load-bearing and some are wise — but to see them as *placed* rather than given. An inherited component can be kept, deliberately, by the person whose board it is. That is a different thing from one that was simply never examined, and the difference is the whole of the exercise.

This is also why the notation and the symbols need each other most here. The notation will tell you the state you are in. Only the schematic will tell you who built it.

2.5 The same expression, two boards, two lives

If the notation reports and the board commits, then the obvious question follows, and the answer is uncomfortable and useful in equal measure.

Two completely different boards can implement the same expression — and produce outcomes that have almost nothing in common.

Read that again with the human case in mind. Two people can have *identical* expressions for an event and live completely different versions of it. Same Noetic, same Domain, same Drive, same Register, same direction of transition. Radically different lives.

What the expression fixes

Not nothing. The expression pins down five things, and they matter:

1. **What kind of event it is** — which Noetic: a trigger, an arrival, retention, repulsion, a rhythm.
2. **Where it sits** — the Domain: principled, cognitive, evaluative, operational.

3. **What is powering it** — the Drive.
4. **How it is experienced** — the Register: mental, emotional, physical, spiritual.
5. **The direction of the transition** — trigger to arrival, crossing to crossing.

That is the **shape** of the event. It is a real constraint. It tells you what class of thing you are looking at, and it rules out whole families of description.

What the board fixes, and the expression cannot

Everything about the event that makes it bearable or catastrophic:

Decision	What it controls
Gain	whether the same trigger lands as irritation or terror
Bias / operating point	what the trigger even means when it arrives
Thresholds	whether anything protective fires at all
Clamps and ceilings	how far the output can go before it stops rising
The return path	whether the loop damps and fades, or sustains and escalates
Time constants	whether it takes a second or three weeks
Isolation	whether the consequence reaches back and moves the control
Defaults and failure states	what happens when the input is unreadable
Reachability	whether the safety element *can* trip, or is decoration

None of those is in the expression. All of them are decisions, and all of them change the outcome more than the expression does.

The same trigger, two boards

Take one expression, held identical in both people:

```
^8^C_{F4}|A  +  ^5^P_{F7}|T   ->   ^9^C_{F4}|f
```

A sound from outside — someone's laugh, a tone of voice, a door — meeting accumulated composition and landing as a felt arrival. The same shape in both.

Board A. Gain high and unclamped. No dead-band on the decision stage, so it chatters. No return path, so nothing ever comes back to inform the next moment. No stated default. The same sound lands as a certainty about the whole relationship, and it lands in a second.

Board B. Gain limited by a resistor. A clamp at 0.85, so the output cannot run to the rail. A dead-band on the decision, so a single sample cannot commit anything. A slow return path with τ measured in hours, so the day's evidence gets to weigh in. A fuse that is reachable, and a stated fail-closed default.

Same expression. In one, the sound is a catastrophe. In the other, it is a pang that passes and gets thought about in the evening. Neither person is confused about what happened. They are running different boards.

Why this matters more than it looks

"We have the same problem" is usually false. Two people compare expressions — *we both get like this when he raises his voice* — and conclude they share a condition. They share a **shape**, and differ on every parameter that determines how bad it gets. The conversation that follows is usually an argument about whose experience is the real one. It is actually two boards being compared as though the expressions were the whole story.

This is why reframing alone often does nothing. Renaming the event changes the *description*. If the gain is still high and there is still no return path, the board runs exactly as before and the new name

is just a better slogan for the same output. Reframing works when it changes a parameter — when it moves a threshold, widens a dead-band, or introduces a return path — and does nothing when it only changes the label.

And it is why board changes work when nothing else does. Sleep is a rail. A stated boundary is a fuse, if it is reachable. Leaving the room at the start rather than the end is a dead-band. Telling one other person is a return path. None of these changes the expression at all — the trigger still triggers, the arrival still arrives — and all of them change the outcome. This is the most hopeful fact in the book: **you can change your life without changing what happened to you, because most of what determines the outcome was never in the event.**

The guardrail on this claim

The expression is not empty, and it would be a mistake to read this section as "anything goes, it is all in the design."

The shape constrains the *class* of outcome. An `^8 -> ^9` transition is an associative landing, and no board will make it into a reasoned conclusion — that requires a `^6` operation, which is a different expression. A `^3` repulsion will not become `^2` attraction through better engineering; that conversion does not exist. Retention will not stop being retention. **You can change the magnitude, the speed, the reversibility and the failure mode. You cannot change what kind of event it is.**

Which is the whole division of labour, stated once more: the notation tells you what you are dealing with, and the board tells you what you have decided to do about it.

2.6 The same experience, written both ways

An example, so the duality is concrete rather than asserted.

The experience. A man is in a shop and a woman's laugh, two aisles over, lands in him as a whole afternoon in 1998 — the light, the room, the fact that he was happy — and he has to put a hand on the shelf.

The expression.

```
^8^C_{F4}|A   ->   ^9^C_{F4}|f

Cause:      an external sound, cognitive domain, Connection, analytical
↪  intake
Substrate:  ^5^P_{F7}|T — thirty years of composition, principled,
↪  Purpose, transcendent
Effect:     an arrival, cognitive, Connection, affective
```

Read the substrate line carefully, because this is what the notation gives you that a paragraph does not. What made the laugh land as a *whole afternoon* is not the laugh. It is `^5` — what the man is made of — shaped in the principled domain and carrying his sense of what his life has been for. The notation refuses to let you write the event as the cause of the feeling, because the event is only the trigger and the landing is supplied by the substrate.

The symbol. One transistor. Cause on the base side, the accumulated substrate on the bias terminal, an arrival at the output. The drawing says the same thing in a shape you can hold in your head: *the same input into a different bias gives a different output*, and no amount of changing the input will fix a bias problem.

Now the enforcement, both ways.

The notation forbids writing `^8 -> ^2` — "the laugh caused happiness." That is not a trigger-to-arrival transition; it is a claim that an external `^0` Idea put `^2` Positive into a Mind, and there is no component for that. Correcting the expression to `^8 … -> ^9` produces the *drawing*: a trigger and an arrival, with the bias doing the shaping.

The drawing forbids a wire that manufactures its own quantity. Correcting the drawing — insisting that the output comes from the substrate — produces the *expression*: `^5^P_{F7}|T` is doing the work,

and if he wants the laugh to land differently, the `^5` is what has to change. Slowly. Over years.

Two representations. One structure. Each one caught what the other would have let through, and the man is better described than he was before either of them was written.

2.7 Why this matters for the rest of the book

Every case study that follows is written in both languages. The prose carries the human situation, because that is what you and I live in. The expressions carry the precision, because human language is very good at hiding an error inside a fluent sentence.

When they disagree, believe the pair, not either one alone. A paragraph that sounds wise and an expression that will not compile is a paragraph that is wrong — and the notation has just saved you from acting on it.

The rest of this book uses four register labels throughout: `|A` mental, `|f` emotional, `|E` physical, `|T` spiritual. Those four words are the notation's own fourth axis, and the reason the whole library can be read as a description of a person rather than a description of a board.

Chapter 3

The Signal and Annotation Legend

Plate So.

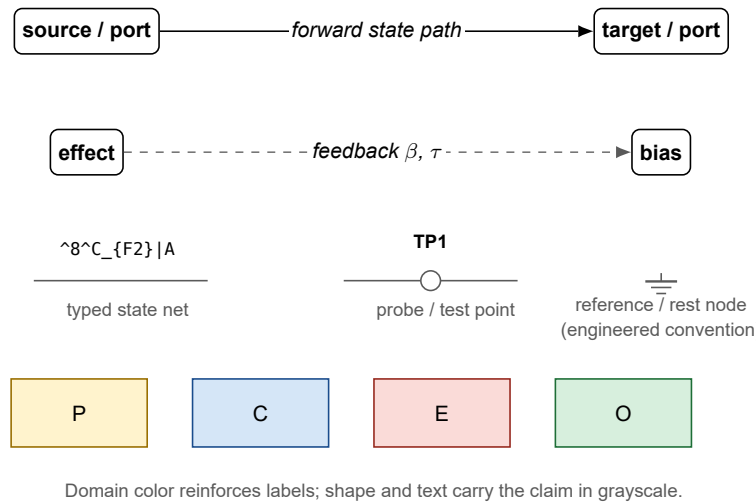


Figure 3.1: The signal-and-annotation legend. Solid arrows move forward; dashed arrows return feedback; every cognitive net is typed or explicitly named.

The legend in one sentence: port, name, direction, return, reference. A port begins or ends a path. A name says what the whole signal is. A solid arrow carries it forward. A dashed arrow brings a sampled piece of it back. A reference fixes the engineered zero.

Read as a person rather than a board, the five parts are the five things you need in order to say anything precise about an experience.

A port is a boundary. *"This starts when she walks in. This ends when I leave the car."* Without boundaries, a description of what happened spreads until it covers the whole week and stops being useful.

A name is what the wire carries, written down. Not *"I was in a mood"* — a mood is not a signal, it is a weather report. *"I was angry with him, and underneath it I was frightened"* names two wires. Naming is not decoration: an unnamed wire cannot be traced, and most of what people call confusion is a bundle of unnamed signals.

A solid arrow is the forward path: this leads to that. *"I did not sleep, and by Thursday I was snapping at people."* Forward paths are what you intervene on if you want to change an outcome.

A dashed arrow is the return. A sampled piece of what came out goes back and changes what goes in. This is the one people miss, and it is why the same argument happens five times: the second argument is not a new event, it is the first one coming back around. On a schematic this is drawn dashed so you can tell the difference at a glance. In a life, the returning signal usually arrives wearing the clothes of a fresh cause, which is exactly why you have to look for it on purpose.

A reference is the zero you measure from — and this is the one that surprises people. A reference is a **convention**, chosen so that other things can be compared. It is not discovered, and it is not sacred. The most useful question anyone can ask about a strongly-held reaction is not *"why do I feel this?"* but *"what am I measuring this against, and who set that zero?"*

Two rules come with the reference, and both matter later:

- **It is not `^0` Idea by identity.** Calling a calm baseline "ground" is an engineering convenience, not a metaphysical claim.
- **A reference can be wrong.** Somebody raised in a house where the atmosphere had to be monitored has a zero point set by that house. By that zero, a quiet room reads as *something is about to*

happen. Nothing is broken except the calibration, and recalibration is a real, slow, possible thing.

Part II

The Parts, and the Lives They Explain

Every symbol, in all four registers

Chapter 4

Flow and State Retention

Plate S1.

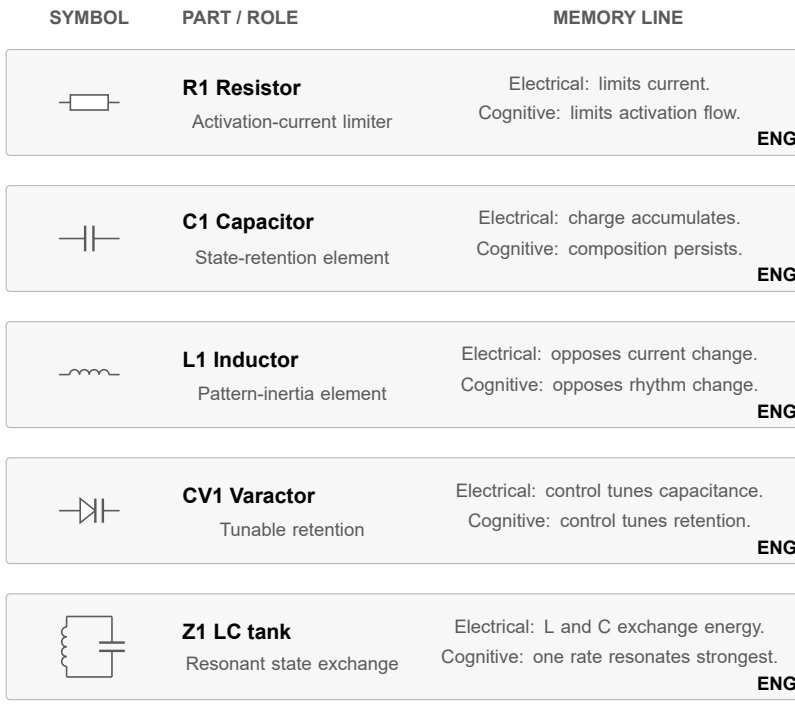


Figure 4.1: Flow and state-retention symbols. The short memory line states the preserved functional role; ENG marks a stipulated cognitive quantity.

Resistor — the part that limits how fast something gets through

The experience. Your brother says something designed to wound you at a family lunch. You feel the whole answer arrive — fully formed, accurate, and lethal — and you say, "Pass the salt." Nothing about the anger changed. The rate at which it reached your mouth did.

The same thing, in all four registers.

- `|A` **Mental** — counting to ten before you answer; the argument you build and then decline to make
- `|f` **Emotional** — not letting someone see how much they have just hurt you
- `|E` **Physical** — the slow breath; the hand that stops halfway to the phone
- `|T` **Spiritual** — restraint kept because of who you intend to be, when nobody would ever know

What is happening. The activation is real and undiminished. What has been added is a limit on flow: the same pressure, delivered slowly enough that the decision about what to do with it happens somewhere other than mid-sentence. In CAS terms this is restraint on propagation. It lives mostly in `|E` — the deliberate half-second of breath — while the judgment it buys is spent in `|A`.

The part that matches. A resistor does not reduce the voltage. It limits current for a given voltage, and it dissipates what it limits as heat. The correspondence is exact at the level of roles: pressure unchanged, rate constrained, cost paid. The cost is real in both cases — the swallowed answer goes somewhere, and it is not free.

Where it breaks. A resistor is linear and passive: it limits everything that passes, always, without judgment. Human restraint is selective and expensive. Someone who limits *every* answer is not well-regulated, they are shut down, and the map stops describing them.

Capacitor — the part that keeps what arrived

The experience. Twenty years after she died, you can still hear the exact way your mother said your name when she was pleased. You did not rehearse it. You did not choose to keep it. It stayed.

The same thing, in all four registers.

- `|A` **Mental** — a fact you can still recall on demand
- `|f` **Emotional** — warmth that colours a room twenty years later
- `|E` **Physical** — the softening in your shoulders at the sound of a voice
- `|T` **Spiritual** — part of what you would name if asked who you are

What is happening. Something arrived once and has persisted, and the persistence is itself doing work — it shapes how you hear praise today. This is `^5` Female: the Mind **is composed of** accumulated Ideas. Note the register: in `|A` it is a fact you can recall on demand; in `|f` it is a warmth that still colours a room; in `|E` it is a particular softening in your shoulders; in `|T` it is part of what you would name if asked who you are.

The part that matches. A capacitor holds charge across time and resists a change in the voltage across it. Accumulated composition persists and can decay. That is the whole of it, and the role-match is exact.

Where it breaks — and this one matters. A capacitor is a **container**. `^5` Female is not. The Mind does not hold experiences the way a bucket holds water; the Mind is made of them. The moment you write "the Mind contains her memory," you have imported a picture the framework explicitly rejects — a self standing behind its own contents. Write "the Mind is composed of it." The distinction is not pedantry: a container can be emptied and still be a container, and a person cannot.

The leak is real too. Capacitance leaks. Good retention is not perfect retention, and the leak rate is a parameter, not a moral failure.

Someone who remembers everything at equal vividness is not healthier than someone whose grief has softened a few percent a year.

Inductor — the part that will not change speed

The experience. You have cracked your knuckles the same way since you were nine. You decide, at forty, to stop. Nothing about the decision is difficult. The hand does it anyway — three times before you catch it — and catching it feels like interrupting a sentence.

The same thing, in all four registers.

- `|A` **Mental** — the reasoning that keeps arriving back at the old conclusion
- `|f` **Emotional** — loyalty to a person or an idea you have already outgrown
- `|E` **Physical** — the hand that cracks a knuckle before you notice
- `|T` **Spiritual** — a faith or a vocation that will not be hurried into changing

What is happening. An established pattern is resisting a *change in its rate*. Not the pattern itself — the change. Inductance opposes any change in current, and it pushes back in proportion to how fast you are asking for the change. This is why habit advice that says "just stop" fails: it requests the highest possible rate of change, and therefore meets the largest possible opposition.

The part that matches. An inductor stores energy in a magnetic field and fights a change in current, and it keeps the current flowing when you open the switch — which is why an inductive circuit arcs. That is not a quirk of the analogy; it is the same statement. Abruptly interrupting an established pattern releases the stored energy somewhere, and if you have not provided a path, it will find one.

Where it breaks. An inductor's opposition is symmetric and instant. Human inertia is neither: it is weaker for a pattern being *increased* than for one being cut, and it has a refractory period. And

unlike a coil, a person can change their own inductance over months — practice reshapes the pattern rather than merely pushing back on its rate.

Varactor — the part whose retention can be tuned

The experience. The smell of a particular brand of cigarettes used to put you straight back in your father's car, and it hurt. Ten years on, the same smell arrives and you think: *him* — and there is more affection in it than pain.

The same thing, in all four registers.

- `|A` **Mental** — reframing: the same fact, carrying less weight
- `|f` **Emotional** — a hurt that softens from agony into tenderness
- `|E` **Physical** — a flinch that becomes a pause
- `|T` **Spiritual** — a loss that slowly becomes a source of meaning rather than a wound

What is happening. Nothing was deleted. This is the register people miss when they ask how to *forget* something: the memory's **retention was retuned by a control input**, not erased. Therapy, time, and a changed relationship to the memory act as that control. The composition is the same; how strongly it holds, and how sharply it swings when it discharges, has changed.

The part that matches. A varactor is a diode whose junction capacitance is set by a reverse-bias voltage. It is not a different component — it is the same junction, tuned by a control signal. That is exactly the human case: same retained content, externally tunable persistence.

Where it breaks. A varactor's control is monotonic, reversible, and smooth over its range. Humans are none of those. The same intervention that softens a memory can harden it, the change can be irreversible in one direction, and the curve is not smooth. Read the varactor as *"retention is controllable"* — never as *"turn the dial to reduce pain."*

LC tank — the part that selects one frequency

The experience. You and your partner can discuss money, in-laws, your careers, and the children without difficulty. There is one subject — who was going to move, twenty years ago — that ends the same way every time. Not a similar way. The same way.

The same thing, in all four registers.

- `|A` **Mental** — the argument you could have in your sleep — the same three points
- `|f` **Emotional** — the subject that always ends the same way
- `|E` **Physical** — the room, the hour, the particular tone that sets it off
- `|T` **Spiritual** — the unresolved question about what it all meant, re-asked every time

What is happening. Coupled retention and inertia, exchanging energy, are picking out one repeating rate and favouring it. The capacitance holds the accumulated charge of the old grievance; the inductance is the established pattern of how the two of you move through it. Together they resonate, and a resonant circuit responds to one frequency far more strongly than to any other. That is why the argument can be set off by almost nothing: a resonant tank barely needs input at the right rate.

The part that matches. An LC tank stores energy, swapping it between the capacitor's electric field and the inductor's magnetic field, and it does this most readily at one frequency set by $f_0 = 1/(2\pi\sqrt{LC})$. The role-match to "one recurring fight, one recurring rhythm" is exact, and it predicts something useful: to change the frequency you must change L or C — the retained charge or the established pattern — not merely resolve to argue better.

Where it breaks. A tank is passive and its Q is fixed by its components. A couple can raise or lower their own Q by learning to notice the moment the tank starts ringing. And unlike a circuit, the human

version usually has *more than one* tank, and the interesting behaviour is in how they couple.

Chapter 5

Direction, Threshold, and Sensing

Plate S2.

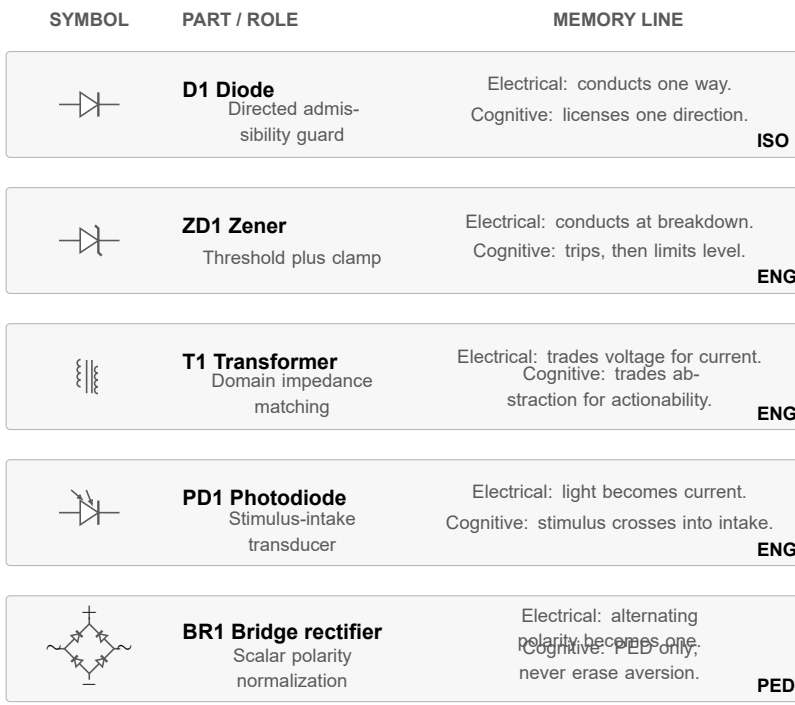


Figure 5.1: Direction, threshold, conversion, and sensing symbols. The bridge rectifier is intentionally limited to a scalar-polarity teaching analogy; it does not convert Negative into Positive.

Diode — the part that only lets things through one way

The experience. You forgive your brother for what he did with the money. You mean it, and it holds. And you notice, two years later, that you no longer lend him anything — not as a policy, and not as punishment. The knowledge simply will not travel back the way it came.

The same thing, in all four registers.

- `|A` **Mental** — knowing something you cannot un-know
- `|f` **Emotional** — forgiveness that is real, and trust that did not come back with it
- `|E` **Physical** — a body that will not relax in that person's presence
- `|T` **Spiritual** — a blessing you can still give, in one direction, to someone you cannot let back in

What is happening. An admissibility relation has been licensed in one direction and refused in the other. Forgiving is the forward path. Forgetting-what-you-learned is the reverse path, and the reverse path is not blocked by stubbornness — it is blocked structurally. This is why "you said you forgave me" and "I still don't trust you with money" can both be true and neither is a lie.

The part that matches. A diode conducts predominantly one way: forward biased, current flows; reverse biased, it does not. It is a one-way gate, not an obstacle. The roles correspond one-for-one, and this is one of the clearest ISO mappings in the library.

Where it breaks. A real diode has a reverse breakdown voltage, and past it the gate conducts — hard, and often destructively. So does a person. There is a pressure past which the thing you swore would never get back through comes through all at once, and it does not come through gently.

Zener — the part that trips at a level and then holds a ceiling

The experience. You can take criticism. You have taken a great deal of it and used it well. Then someone says the one thing — about your parenting, about your father — and something changes: your voice flattens, you become extremely polite, and you say, "That's fair, I'll think about it." Nothing more gets in. You are not angry. You are clamped.

The same thing, in all four registers.

- `|A` **Mental** — the smooth, agreeable sentence that arrives once you are past your limit
- `|f` **Emotional** — going flat and warm and unreachable under one specific criticism
- `|E` **Physical** — the face that stops moving; the voice that levels out
- `|T` **Spiritual** — protecting a core commitment instead of arguing about it

What is happening. Below the threshold the channel is closed and criticism passes through your ordinary processing. At the threshold, a different path conducts, and its job is to cap the output. The politeness is not a strategy; it is the regulator doing its only job, which is to stop the level rising further. In register terms: `|f` has hit a ceiling, and `|A` takes over and produces smooth, agreeable sentences.

The part that matches. A Zener diode blocks in reverse until its breakdown voltage, then conducts and clamps the voltage there. Two distinct behaviours, one component, and the transition is sharp rather than gradual. The correspondence holds, including the sharpness — people do not degrade gracefully into their clamp.

Where it breaks. A Zener's breakdown voltage is a stable datasheet number. A person's is not: it moves with sleep, with who is in the room, and with how the last hour went. And a Zener clamps cleanly. A human at the clamp is still doing something underneath,

and telling someone they are "being defensive" misses that the clamp is protecting a specific, identifiable thing.

Transformer — the part that changes level without changing content

The experience. You explain the same idea three times in one day: to your eight-year-old at breakfast, to a colleague at eleven, and to your mother on the phone at nine. Same idea, three levels, three vocabularies — and the eight-year-old's version is not a lesser version of the colleague's. It is the same content at a different impedance.

The same thing, in all four registers.

- `|A` **Mental** — the same idea expressed at three levels of abstraction
- `|f` **Emotional** — saying a hard thing gently without removing the hardness
- `|E` **Physical** — pitch, pace and posture adjusted to whoever is in the room
- `|T` **Spiritual** — carrying a meaning across a gulf — to a child, a culture, or a generation

What is happening. A licensed crossing is happening, and the converter is trading abstraction against actionability without adding or removing what is being said. The design decision is the turns ratio: how much precision you spend to get how much grip.

The part that matches. A transformer trades voltage against current by a turns ratio. It changes *level* and *impedance*. It does not change content, and — the part that makes this a real correspondence rather than a rhyme — it provides **isolation**: the two sides are not electrically connected. Explaining something to a child and explaining it to a colleague can genuinely be separate conversations, and neither contaminates the other.

Where it breaks. A transformer is lossy and its ratio is fixed at manufacture. A person's ratio is adjustable in the moment and improves with practice, and the failure mode is not loss but distortion — usually by flattering the listener, which adds something the original did not contain. A transformer never flatters.

Photodiode — the part that lets the world in

The experience. You have no particular religious belief and no expectation of anything, and you walk into a cathedral you were only visiting for the architecture — and your eyes fill. Or you are driving and a piece of music starts, and by the second bar you have pulled over.

The same thing, in all four registers.

- `|A` **Mental** — a new fact that changes what you thought you knew
- `|f` **Emotional** — being moved by music, a building, or a face
- `|E` **Physical** — the hair lifting; the breath catching
- `|T` **Spiritual** — awe — an encounter with something you did not make

What is happening. Content arrived from outside. You did not generate it and you did not choose it. The precise reading matters here: `^0` Idea is the content — everything in and outside the Mind — and `^1` Mind is the operation that intakes it. The photodiode does not contain the light and does not invent it. It transduces an external stimulus into something the circuit can use.

The part that matches. A photodiode converts incident light into current. The stimulus is external; the device is a transducer, not a source. The two roles map cleanly: external content in, internal signal out, and the operation belongs to the device while the content belongs to the world.

Where it breaks. It is tempting — and wrong — to say the photodiode's output *is* the light, or that a person's response *is* the stimulus.

The transducer shapes what it admits: its sensitivity curve, its bias, its angle. Two people in the same cathedral are not receiving the same signal, and neither of them is receiving the building itself. Never write `Photodiode< stimulus -> ^0 >`. The output is a `^1` intake, and the `^0` stays on the far side of the boundary.

Bridge rectifier — the part that turns a swinging scalar into a magnitude

The badge is PED here, and the reason is worth reading.

The experience. You wake up in a mood that has no direction. You are not sad and you are not happy; you are activated. Your skin is prickly, your thoughts are fast, and you cannot sit down. Friends ask what is wrong. Nothing is wrong. Something is *up*.

The same thing, in all four registers.

- `|A` **Mental** — realising you cannot say whether the day was good or bad, only that it was loud
- `|f` **Emotional** — keyed up with no direction; arousal without valence
- `|E` **Physical** — restlessness, prickling skin, no target
- `|T` **Spiritual — no honest reading.** This is why the badge is PED. `|T` cannot be manufactured from raw activation

What is happening. A quantity that usually has a sign has been read as a magnitude. This is a real and useful description — arousal without valence is a genuine state, and naming it as a magnitude rather than searching for a missing emotion is often what unsticks someone.

The part that matches, within limits. A bridge rectifier turns an alternating input into one direction of output. If — and only if — the input is explicitly defined as a **non-valence scalar**, such as `signed_error -> magnitude`, this is a legitimate engineering structure. The mapping is pedagogical: it teaches the shape of the operation.

Where it breaks, and this is the hard rule. A bridge rectifier must **never** be used to turn `^3` Negative into `^2` Positive. These are not two signs of one axis. `^3` is repulsion and `^2` is attraction; they are an involution pair, not a polarity flip, and there is no circuit that converts one into the other. Anyone who offers you a component that turns hatred into love has handed you a category error with a schematic attached. If your diagram needs that conversion, the diagram is wrong.

Chapter 6

Active and Comparative Parts

Plate S3.

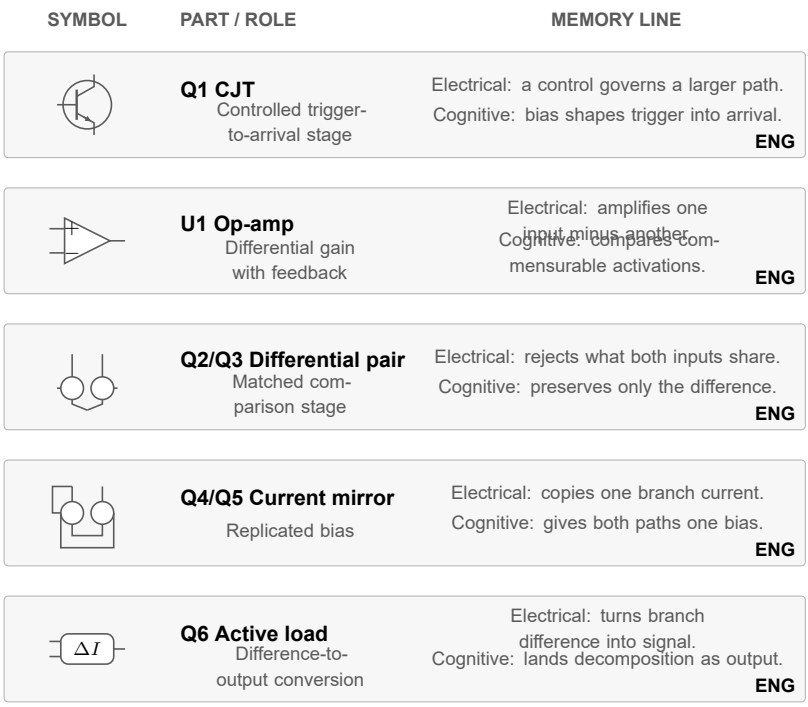


Figure 6.1: Active and comparative parts. Cause, Bias, and Effect are cognitive terminal roles on the CJT; they are not claims that a standard BJT's collector, base, and emitter have those signal roles in every configuration.

CJT — the part where what arrived depends on what you are made of

The experience. Two people hear the identical sentence — *"we should probably talk about your mother"* — in the same kitchen, at the same volume. One of them feels a small administrative irritation: *fine, let's get it over with.* The other's stomach drops and they are nine years old again. Same input. Different output. Nothing is wrong with either of them.

The same thing, in all four registers.

- `|A` **Mental** — one sentence, two people, two entirely different conclusions
- `|f` **Emotional** — the same remark landing as mild irritation in one and a wound in the other
- `|E` **Physical** — the stomach dropping before any thought has formed
- `|T` **Spiritual** — a sentence that lands as calling, because of the life it meets

What is happening. This is the central component of the whole system, and it is worth slowing down for. An `^8` Above/Cause — an Idea that triggers a *different* Idea than itself to come to Mind — meets a `^5` Female substrate, the accumulated composition that the Mind is made of, and what lands is an `^9` Below/Effect. The trigger does not contain the response. The substrate supplies it. That is why the same sentence produces grief in one person and logistics in another, and why telling someone to "be reasonable" about it does not work: you are asking them to change their substrate by an act of will, in the middle of the moment it is doing its job.

The part that matches. A transistor is the same device everywhere it appears, and its behaviour is set by its bias. Change the bias and the identical input produces an entirely different output — cut-off, or amplification, or saturation. The role correspondence is exact: Cause enters, Bias shapes, Effect lands.

Where it breaks, and this is the anti-drift rule for the whole book. `^8` Above/Cause is **associative triggering**, not reasoning. Its colloquialisms are "that reminds me," "one thing leads to another," "it triggered a memory." If you find yourself describing someone as *analysing* the situation, or *reasoning* their way to a conclusion, you are no longer describing `^8` — you are describing a compound of `^6` Male (decomposition) and `^9` (the landing). Getting this wrong is the single most common error in this framework, and it produces advice that sounds sophisticated and helps nobody.

Op-amp — the part that amplifies a difference between two things

The experience. Your partner says "nothing's wrong" in a voice that is a quarter-tone flatter than usual. By the time you have finished washing up, you have constructed an entire theory of the last three weeks from that quarter-tone. You were not wrong that something was different. You were wrong about how much.

The same thing, in all four registers.

- `|A` **Mental** — building a three-week theory out of one flat tone of voice
- `|f` **Emotional** — one cool goodbye, and a certainty that you are not loved
- `|E` **Physical** — hearing a quarter-tone in a voice before you know you heard it
- `|T` **Spiritual** — reading a small event as a verdict on your whole direction — usually a gain error

What is happening. A small difference between two quantities was amplified enormously because there was no feedback path holding the gain down. In CAS terms, a tiny difference in `|E` — a voice, a posture — was taken up and run through `|A` until it became a certainty. The input was real. The magnitude was manufactured.

The part that matches. An op-amp takes the difference between two inputs and multiplies it by an open-loop gain so large that the output is meaningless without feedback. The minus input is exactly that: **arithmetic subtraction of two commensurable quantities.** And the discipline this predicts is real — you must close the loop with feedback (ask the person directly) or the output will be driven to a rail by noise.

Where it breaks — an important precision. An op-amp's inverting input is arithmetic subtraction. It is **not** a Noetic involution. Involution is defined only for `^2↔^3`, `^5↔^6`, and `^8↔^9`; the self-duals

`^0`, `^1`, `^4` and `^7` remain themselves. Writing "the op-amp inverts the state" and meaning "turns Positive into Negative" is a category error. Inverting an *arithmetic* difference around a reference is ordinary engineering and has nothing to do with the involution structure.

Differential pair — the part that ignores what both sides share

The experience. You and a colleague are made redundant on the same morning, from the same meeting, with the same words. Eighteen months later you are both working again and both broadly all right. But you notice something: your colleague was angry for a fortnight and then fine, and you were calm for a fortnight and then fell apart in the third. Same event. The difference between you is the part that was actually about you.

The same thing, in all four registers.

- `|A` **Mental** — two people given identical news, reaching different conclusions; the difference is them
- `|f` **Emotional** — the same bereavement, two different depths of grief
- `|E` **Physical** — the same cold room, and only one of you shivers
- `|T` **Spiritual** — the same loss, and one person's faith holds while another's does not

What is happening. The shared circumstance is **common-mode**, and a well-built comparison cancels it. What remains is the difference — the part that is not explained by the event, and therefore has to be explained by the person. This is uncomfortable precisely because it is informative: it removes the excuse and leaves the signal.

The part that matches. A differential pair with good common-mode rejection amplifies only the difference between its two inputs. Matched devices and a shared bias are what make the rejection work. Unmatched inputs — two things that are not actually comparable

— produce garbage, which is the technical statement of "you can't compare those two situations."

Where it breaks. A differential pair needs **matched** devices. Comparing your reaction to a redundancy with your reaction to a bereavement is an unmatched comparison, and it will output noise that you will then mistake for insight. State the axis before you compare anything.

Current mirror — the part that copies one condition onto another branch

The experience. You are forty-five and you catch yourself doing the exact thing your father did when a bill arrived: not opening it, moving it to the side of the desk, feeling vaguely sick. You have never once decided to do this. And you are doing it to a piece of paper in your own house, thirty years after leaving his.

The same thing, in all four registers.

- `|A` **Mental** — an inherited way of reasoning you never chose and never noticed
- `|f` **Emotional** — your mother's dread arriving in your chest at an airport
- `|E` **Physical** — a gesture, a posture, a way of holding a cup
- `|T` **Spiritual** — an inherited faith, or an inherited cynicism

What is happening. A control condition was **copied**, not chosen. This is one of the most useful things the library can describe, because it explains behaviour that has no decision behind it and therefore cannot be fixed by deciding differently. The bias was replicated onto your branch; the branch itself is your own.

The part that matches. A current mirror replicates the current in one branch into another by tying the two control terminals together. The second branch did not compute anything; it received the same bias. The role-match is clean and — importantly — it comes with the

fix: you can change the copied current only by changing the **reference** branch, which in human terms means working on the pattern where it lives, not on your own symptoms.

Where it breaks. A mirror copies continuously and exactly. Human transmission is partial, shaped by temperament, and interrupted by everything that happened to you that did not happen to them. You are not a copy of your father with a wire attached. You are a branch that shares one control node with him, and the rest of the circuit is yours.

Active load — the part that turns a difference into an actual decision

The experience. You have felt vaguely off at this party for an hour. Nothing is wrong, exactly. Then someone makes a small remark, and you think: *I'm leaving.* The feeling was there the whole time. What changed was that it became an output.

The same thing, in all four registers.

- `|A` **Mental** — the moment a vague suspicion becomes a decision
- `|f` **Emotional** — unease turning into resolve
- `|E` **Physical** — standing up and reaching for your coat
- `|T` **Spiritual** — a long-felt sense that something is wrong becoming a commitment to change a life

What is happening. A branch difference that was sitting as a difference has been converted into a single-ended output with enough drive to do something. This is the stage that turns a sensed difference into an action, and it is a **composite** — a combination of parts doing a job — not a new primitive. In register terms, `|E` produced the difference and the conversion is what allowed `|O` — operational, the domain of action — to take it up.

The part that matches. An active load replaces a passive resistor with an active device, which raises gain and allows a differential

stage to produce a usable single-ended output. The role-match is that this is a conversion stage: difference in, drive out.

Where it breaks. Do not reify it. An active load is not a faculty, a module, or a homunculus that decides things. It is two transistors arranged so that a difference arrives somewhere it can be used. The moment you give it agency you have invented a little person inside the person, which is exactly the move this library exists to prevent.

Chapter 7

Rhythm and Register Conversion

Plate S4.

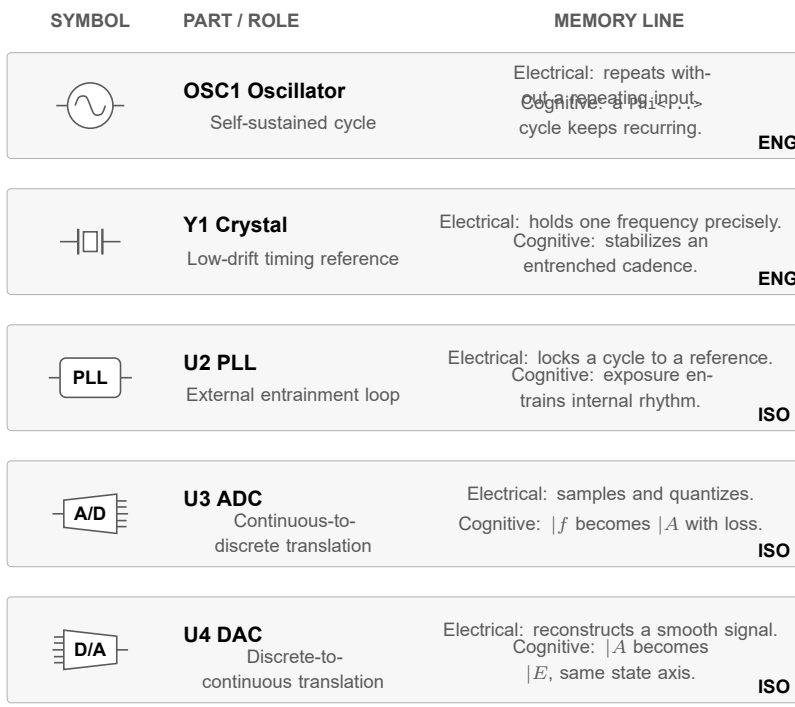


Figure 7.1: Rhythm and conversion parts. Pure ADC/DAC register examples keep Noetic and Domain fixed; a Domain change requires a separate directed crossing or conversion stage.

Oscillator — the part that runs on its own

The experience. It is 2 a.m. and you are going over it again. Not solving anything — going over it. The same four minutes of last Tuesday, the same sentence, the same imagined reply. You have been awake for an hour and a half and you have not produced one new thought.

The same thing, in all four registers.

- `|A` **Mental** — rumination — the loop that produces no new thought
- `|f` **Emotional** — a mood sustaining itself without new cause
- `|E` **Physical** — pacing; the body keeping the loop running
- `|T` **Spiritual** — a question about meaning that returns every night and is never answered

What is happening. A loop is sustaining an ordered cycle with no fresh input. That last part is what makes it an oscillator and not a process: it is not waiting for more information, because it does not need any. Positive feedback has taken over, and the output has become the input. This is the structure of rumination, and it explains why it feels like thinking while producing nothing.

The part that matches. An oscillator uses positive feedback to sustain a periodic signal with no external drive. The role-match is exact, and it comes with the engineering answer: you stop an oscillator by adding loss, by breaking the feedback path, or by capturing it with something stronger. In human terms: change the environment, interrupt the loop physically, or give it a competing rhythm.

Where it breaks. An oscillator's frequency and amplitude are stable and predictable. Rumination is not: it wanders, it escalates, and it is sensitive to everything from blood sugar to who texted you. Treat the oscillator as the *shape* of the loop, not a claim about its timing.

Crystal — the part that holds a steady beat

The experience. There is a man in your village who has walked the same route at the same hour for thirty years. You have never spoken to him. On the two occasions he missed it — a funeral, and the week of the storm — several people noticed and asked after him. His steadiness is load-bearing for people it has never occurred to him he was holding.

The same thing, in all four registers.

- `|A` **Mental** — a principle you apply the same way every time
- `|f` **Emotional** — warmth that is the same on a bad day as a good one
- `|E` **Physical** — a daily practice kept at a fixed hour
- `|T` **Spiritual** — integrity — a reference other people set their clocks by

What is happening. A high-`Q` reference: low drift and low pullability. The useful property is not that it is precise in the abstract, but that **other things can lock to it**. A person whose behaviour is the same across contexts — who is recognisably himself at work, at home, and under pressure — becomes a reference other people set their clocks by. That is a real function in any group, and it is usually invisible to the person performing it.

The part that matches. A crystal resonator has a very high Q: it barely drifts, and it is hard to pull off frequency. It is used as a timing reference precisely because it will not be moved by the rest of the circuit. The correspondence holds — including the cost, which is that a very high-Q reference is also very hard to adjust when it *is* wrong.

Where it breaks. A crystal is passive and mindless. Human steadiness is a maintained achievement, it costs something, and it can be exhausted. Someone who has been the reference for everyone for twenty years may be running on no reserve at all — which is the entire subject of Chapter 16.

PLL — the part that catches someone else's rhythm

The experience. You are walking beside someone and, without deciding to, you are in step. A month into a new job, you notice you have started taking breaks when your team takes breaks, at a pace you would not have chosen. A baby's sleep schedule gradually becomes the household's schedule, and then the household's schedule becomes the baby's.

The same thing, in all four registers.

- `|A` **Mental** — absorbing a team's way of thinking in a month
- `|f` **Emotional** — moods synchronising with a partner's
- `|E` **Physical** — falling into step; a sleep schedule entraining the household
- `|T` **Spiritual** — taking on a community's sense of what matters

What is happening. An internal rhythm has **locked** to an external reference. This is `^7` Rhythm: the rate and pattern a Mind is exposed to, entraining it. Crucially, locking is *gradual* and has a **range** — you can only be captured by a rhythm close enough to the one you already keep.

The part that matches. A phase-locked loop compares its own oscillator to a reference, feeds the difference back, and pulls its oscillator into step. It has a **capture range** (how far off the reference can be for lock to be acquired at all), a **lock range** (how far it can drift once locked and stay locked), and a loop time constant. All three have human counterparts, and the prediction they generate is practical: you cannot entrain someone by asking them to change their rhythm to something far from their own. You have to move it in steps that stay inside their capture range.

Where it breaks. A PLL locks to whatever its reference does, including a bad reference, and it does so without judging. So does a person — which is why the interesting question is never "can you entrain?" but "entrained to what, and who chose it?"

ADC — the part where a continuous feeling becomes a name

The experience. For three weeks you have known that something is wrong and have not been able to say what. You describe it as tired. Then, in the middle of an unremarkable conversation, you say out loud — *"I think I'm lonely"* — and something shifts. Nothing about your circumstances has changed. Something has become legible.

The same thing, in all four registers.

- `|A` **Mental** — finding the word for what you have been feeling for three weeks
- `|f` **Emotional** — the feeling becoming nameable, and therefore bearable
- `|E` **Physical** — noticing the clenched jaw and calling it tension
- `|T` **Spiritual** — naming a nameless longing as a question of meaning

What is happening. A continuous `|f` impression has been converted into a discrete `|A` description. This is one of the most valuable operations available to a person: the analytical register can hold, compare, plan, and communicate, and it cannot do any of that with an unquantised feeling. Naming is not a substitute for feeling. It is a translation that makes the feeling usable by the part of you that talks.

The part that matches. An ADC samples a continuous signal and quantises it into discrete codes. In a pure register example the Noetic and the Domain stay fixed — `^4^E_{F2}|f -> ^4^E_{F2}|A` — and only the Register changes. The role-match is exact.

Where it breaks — and this is the part people get wrong. Quantisation is **lossy**. The code is not the signal, and the word is not the feeling. Someone who has become fluent at naming their states can start living in the names, and a name is a much coarser thing than the state it stands for. There is also a resolution limit: below a certain granularity, "fine" and "good" and "okay" are the same code, and using them will make two different weeks look identical.

DAC — the part where a decision becomes a way of living

The experience. You decide to be honest with your father. That is a sentence, and it takes a second. What follows is four years of small continuous behaviour: the phone calls you make when you would rather not, the questions you ask, the thing you do not say at Christmas. The decision was discrete. The living of it is not.

The same thing, in all four registers.

- `|A` **Mental** — a decision becoming a plan
- `|f` **Emotional** — a commitment showing up as daily kindness
- `|E` **Physical** — the decision rendered as habit, posture and routine
- `|T` **Spiritual** — a conversion becoming a life's work

What is happening. A discrete `|A` control has been rendered into continuous `|E` execution. This is the step where intentions either become behaviour or stop, and the gap between the two is where most self-improvement fails. Smoothing matters: the code is stepped, and life is continuous, so the rendering has to interpolate. Somebody who renders "be honest" as a rule rather than a way of moving will be brittle at exactly the moments that matter.

The part that matches. A DAC takes discrete codes and reconstructs a smoothed continuous signal. In a pure register example the Noetic and Domain stay fixed and only the Register changes: `^9^0_{F5}|A -> ^9^0_{F5}|E`. The role-match is the conversion of a stepped specification into continuous output.

Where it breaks — and this one is a hard rule. A DAC is **not** the inverse of an ADC. The ADC is lossy: it quantised a continuous signal into codes and threw away the detail in between. You cannot run the lossy conversion backwards and recover what was lost — no converter can. Anyone who tells you that naming a feeling precisely enough will give them the feeling back has misread the datasheet. The DAC is a separate, forward direction: control in, behaviour out.

Chapter 8

Logic and Routing

Plate S5.

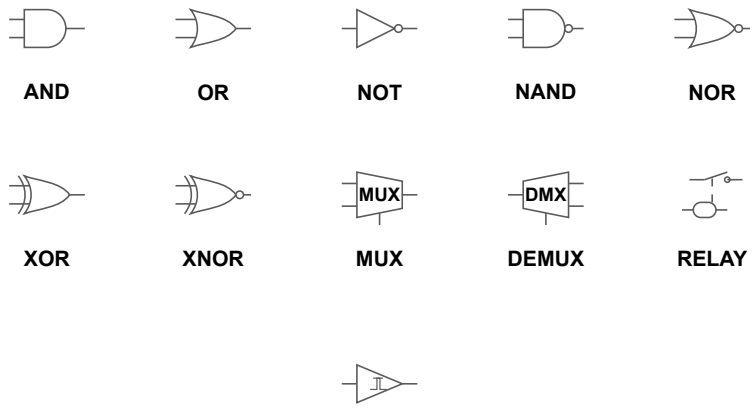


Figure 8.1: Logic and routing symbols. Boolean truth rules apply only after continuous CAS activations have been converted into explicit firing flags.

Gates act on **explicitly thresholded flags** — a continuous state has first been converted into a yes or a no by a stated rule. That prelim-

inary step is where nearly all the argument about these components belongs, because the gate itself is trivial and the threshold is not.

AND — the part that needs every condition at once

The experience. You will lend your brother money if he has a plan **and** he has stopped drinking. Either one alone is not enough, and you know from experience that treating either one alone as enough is how the last two loans went.

The same thing, in all four registers.

- `|A` **Mental** — two conditions that must both hold before you act
- `|f` **Emotional** — love and trust as separate flags; either one alone is not enough
- `|E` **Physical** — calm body **and** willing mind — both, or you will not sleep
- `|T` **Spiritual** — conviction together with practice

What is happening. A decision is being held behind two conditions that must both be active. The useful discipline is that you named them in advance, in the calm, rather than negotiating them in the moment when one of them is failing.

The part that matches. An AND gate is true only when every input is true. Threshold each condition first, and the gate does the rest.

Where it breaks. A well-formed promise may pull you toward helping *and* away from it at the same time. If both flags are genuinely active, the correct reading is not that one of them is fake — it is that you have a **conflict**, and the component for that is AND reporting both-active, not a choice made by shouting louder. Route the conflict to somewhere it can be deliberated. Do not suppress it.

OR — the part where any one thing is enough

The experience. There are three signs that mean you should leave the party: your chest tightening, the third drink, and the moment you start rehearsing what you will say in the car. Any one of them is enough. You do not need all three, and waiting for all three is how you end up staying until two in the morning.

The same thing, in all four registers.

- `|A` **Mental** — any one of three reasons is enough to stop
- `|f` **Emotional** — any one of several feelings means leave
- `|E` **Physical** — any one physical warning sign
- `|T` **Spiritual** — any one thing that breaks a commitment

What is happening. Several flagged conditions are being combined so that any one of them can trigger the exit. This is the structure of a well-designed personal rule: it fires early, it is unambiguous, and it does not require you to argue with yourself.

The part that matches. An OR gate is true when at least one input is true. The role-match is exact.

Where it breaks. OR preserves alternatives; it does not integrate them. Three warning signs do not add up to a fourth, worse thing — they are three separate routes to the same flag. If you want them combined into a severity, that is an arithmetic sum, not a gate, and you should say so.

NOT — the part that flips one flag

The experience. Someone asks whether you are angry. You say no. You are not lying and you are not angry — but you are also not *not* angry in the ordinary sense, and both of you can hear it.

The same thing, in all four registers.

- `|A` **Mental** — "I'm not angry" as a threshold statement, not a fact

- `|f` **Emotional** — equanimity misread by others as coldness
- `|E` **Physical** — stillness that is not calm
- `|T` **Spiritual** — doubt that is not disbelief

What is happening. A flag has been complemented inside a stated frame. The frame is what matters: "not angry" means *below the threshold I am using for angry*, and your threshold and theirs may differ. Naming the frame is what turns a confusing exchange into a clear one.

The part that matches. NOT is the Boolean complement of one input. It is the simplest and most easily abused component in the library.

Where it breaks — hard rule. NOT maps to a Noetic involution only for a licensed pair: `^2`↔`^3`, `^5`↔`^6`, `^8`↔`^9`. The self-duals `^0`, `^1`, `^4` and `^7` are unchanged by it. A Boolean complement of a flag is not the involution of a state, and treating "not attracted" as "repelled" is a category error that will produce bad advice at speed.

NAND — the part that stops when everything lines up

The experience. A safety rule at work: the machine runs unless *both* the guard is open and the operator is clear — at which point it stops. It is the inversion of an all-required condition, and it is written that way because "stop unless it is obviously fine" is a safer default than "go if it looks all right."

The same thing, in all four registers.

- `|A` **Mental** — the one combination that must never both be true
- `|f` **Emotional** — exhaustion plus resentment: stop
- `|E` **Physical** — pain plus pushing through
- `|T` **Spiritual** — a compromise that would defeat the entire point

What is happening. The negation of an AND. Where AND asks *have all conditions been met?*, NAND asks *is this the one combination I must never allow?*

The part that matches. NAND is false only when every input is true. It is the standard building block of fail-safe logic, and its human use is the same: a small number of conditions whose simultaneous presence is unacceptable.

Where it breaks. NAND needs the same explicit thresholds as AND, and it inherits AND's blind spot: it is silent about every combination except the one it forbids.

NOR — the part that acts only when nothing is wrong

The experience. You will have the difficult conversation tonight if nobody is drunk, nobody has just been criticised, and the children are asleep. Not if one of those is fine — if *all* of them are quiet. The rule is deliberately restrictive, and it means some nights the conversation does not happen.

The same thing, in all four registers.

- `|A` **Mental** — act only when every condition is quiet
- `|f` **Emotional** — have the conversation only when nobody is raw
- `|E` **Physical** — only when rested, fed and well
- `|T` **Spiritual** — only when the motive is clean

What is happening. A permissive condition that requires **every** input to be inactive. This is the fail-closed pattern, and its cost is real: it will refuse to act on nights when acting would have been fine.

The part that matches. NOR is true only when every input is false. Use it when the cost of a bad attempt is much higher than the cost of waiting.

Where it breaks. A NOR gate can be starved into permanent silence by one input that never goes false — one person who is never sober, one issue that is never calm. If a NOR condition has been unmet for months, that is not patience; it is a circuit that will never fire, and the remedy is upstream.

XOR — the part where exactly one can pass

The experience. Two people have each told you something in confidence, and you have discovered that telling either one what the other said would hurt them both. You can honour one confidence or the other. You cannot honour both, and choosing both is the same as choosing neither.

The same thing, in all four registers.

- `|A` **Mental** — two conclusions, only one of which can be acted on
- `|f` **Emotional** — two loyalties, one choice
- `|E` **Physical** — two commitments at the same hour
- `|T` **Spiritual** — two callings; you cannot serve both

What is happening. Exactly one of two mutually exclusive paths may be taken. This is a real and common human structure — loyalty conflicts, mutually exclusive commitments, two jobs — and naming it as *exactly one* is often the thing that ends a week of agonising.

The part that matches. XOR is true when exactly one input is true. Standard, unambiguous, and often misremembered.

Where it breaks — and this is the most common error in the whole library. XOR is **not** a both-active conflict detector. When both attraction and aversion are firing, XOR is **false** — it says nothing useful. That situation is AND's job. If you find a diagram using XOR to "detect ambivalence," the diagram is wrong and will mislead you at exactly the moment you need it.

XNOR — the part that reports agreement

The experience. You and your partner both do not want to go to the wedding. Neither of you has said so. When one of you finally does, the relief is instantaneous, because the flags agreed.

The same thing, in all four registers.

- `|A` **Mental** — agreement — both think it, or neither does

- `|f` **Emotional** — mutual indifference mistaken for peace
- `|E` **Physical** — two bodies relaxing at the same moment
- `|T` **Spiritual** — shared conviction

What is happening. Equality between two flags is being reported. Note what that includes: **both-inactive counts as agreement.** Two people who both do not care are in agreement, and a detector that reported only both-active agreement would miss the more common case.

The part that matches. XNOR is true when its inputs are equal.

Where it breaks. Because both-inactive reads as agreement, XNOR alone cannot tell you whether anything is happening. Add an activity qualifier when you need to separate "we agree to do this" from "neither of us cares."

MUX — the part that chooses which response reaches the output

The experience. You have a version of yourself for your mother, a version for your team, and a version for your oldest friend. You do not decide which one to be. Who you are with selects it, and it is selected before you have finished saying hello.

The same thing, in all four registers.

- `|A` **Mental** — which argument you reach for depending on who is asking
- `|f` **Emotional** — the version of your warmth that appears for family
- `|E` **Physical** — accent, posture and volume changing with the room
- `|T` **Spiritual** — which self you present in a place of worship and a place of work

What is happening. Several prepared responses are available and a **named control signal** chooses which one reaches the output.

The control is not hidden: it is whoever is in the room, and naming it is what makes the choice visible and therefore changeable.

The part that matches. A multiplexer routes one of several inputs to one output, chosen by a select line. The role-match is exact, and it predicts the failure: if the select line is stuck, only one response will ever appear.

Where it breaks. A MUX has a defined behaviour for every select value, including a safe default. Human switching does not — under stress, the select line can float and you get whichever version was most recently used, which is rarely the one you would have chosen.

DEMUX — the part that sends one thing to one place

The experience. You have one real insight about how you avoid conflict — and the question is whether it reaches your marriage, your work, or your relationship with your father. It is one insight. It has to be routed.

The same thing, in all four registers.

- `|A` **Mental** — one insight routed to one problem
- `|f` **Emotional** — one feeling allowed into one relationship
- `|E` **Physical** — one habit applied in one setting
- `|T` **Spiritual** — a conviction expressed in one domain of life

What is happening. One input is being dispatched to one of several output paths under a stated control. The routing decision is a decision, and it is usually made implicitly and badly.

The part that matches. A DEMUX sends one input to one selected output. The role-match is exact, and the discipline it teaches is that only the selected output receives the signal — you cannot half-send an insight to two places and expect it to land in either.

Where it breaks. Real life frequently needs the same insight in three places at once, and a DEMUX cannot do that. If you need fan-out, say so and use a different structure — do not pretend the routing was a choice when it was actually a shortage.

Relay — the part where a small signal switches something enormous

The experience. You say one sentence — *"I'm going to hand in my notice"* — and a career, a mortgage, and a set of friendships change state. The sentence was small. What it switched was not. And the two are deliberately kept apart: saying the sentence does not by itself touch the money.

The same thing, in all four registers.

- `|A` **Mental** — one sentence that changes a career
- `|f` **Emotional** — "I'm leaving" — and everything changes
- `|E` **Physical** — a small movement with a large consequence
- `|T` **Spiritual** — a vow that reorganises a life

What is happening. A low-level `^8` Cause control is gating a separate, high-stakes `^9` Effect path, and the two are **isolated** from each other. That isolation is the whole point. It means the small decision can be made in a calm moment without the large consequence reaching back and interfering with the deciding.

The part that matches. A relay uses a small coil current to switch a much larger, electrically separate circuit. The role-match is exact: `Relay< control=^8^E_{F5}|A, switches=^9^O_{F5}|E >`.

Where it breaks — the direction is not reversible. Cause controls Effect, never the reverse. If your diagram shows the large path reaching back and moving the control signal, it is not a relay; it is a feedback loop, and it should be drawn as one. Getting this wrong is how people end up believing that circumstances decide for them.

Schmitt trigger — the part that commits and then holds

The experience. You have decided to end the relationship. You decided it on Tuesday and you were certain. On Wednesday you were certain of the opposite. On Thursday you were certain again, and by

Friday you had re-decided eleven times and told nobody, because the decision would not hold still for long enough to act on.

The same thing, in all four registers.

- `|A` **Mental** — deciding once instead of hourly
- `|f` **Emotional** — holding a decision through a bad night
- `|E` **Physical** — steady hands because the decision is settled
- `|T` **Spiritual** — a commitment that does not wobble with mood

What is happening. You are sitting exactly on a threshold with a little noise on top of it, and the output is following the noise. The fix is not more certainty. The fix is a **dead-band**: a higher level at which the decision turns on and a lower level at which it turns off, with everything in between holding the previous answer. Deciding once, deliberately, with a gap wide enough to absorb a bad night, is a different act from deciding continuously.

The part that matches. A Schmitt trigger switches on at one threshold and off at a lower one, and the gap — the hysteresis — stops the output chattering when the input sits near the boundary. The role-match is exact, and it is one of the most directly useful mappings in the library.

Where it breaks. Hysteresis is a commitment, and commitments can be wrong. A very wide dead-band stops you chattering and also stops you noticing that the situation has genuinely changed. Name both thresholds, and name what evidence would move the lower one.

Chapter 9

Regulation, Protection, and Annotations

Plate S6.

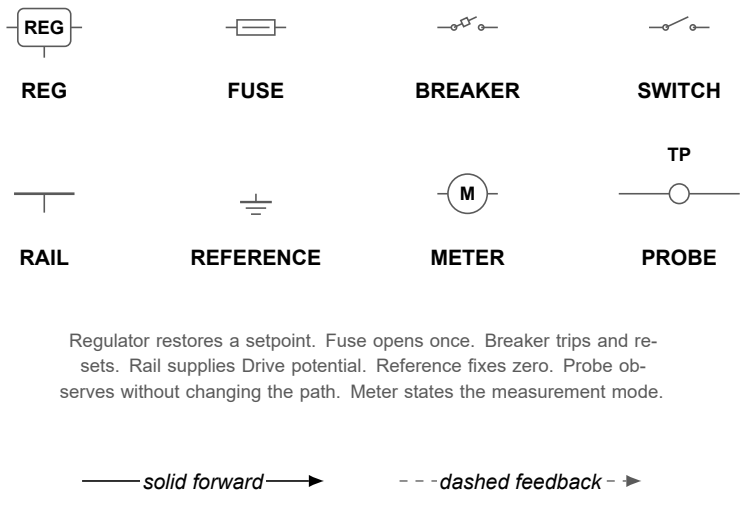


Figure 9.1: Regulation, protection, and annotation symbols. A fuse threshold must be below every upstream clamp or saturation ceiling that feeds it, or the fuse is unreachable.

These are the parts that keep a person running, stop a person running, and let a person see what is going on. They are less glamorous than amplifiers and matter considerably more.

Regulator — the part that restores you to a baseline

The experience. You sleep badly for a week and you notice the change in yourself on the fourth day: shorter fuse, less patience, a tendency to read neutral emails as hostile. You sleep properly for two nights and you are back. Not transformed — back.

The same thing, in all four registers.

- `|A` **Mental** — attention restored by two nights of sleep
- `|f` **Emotional** — mood returning to baseline
- `|E` **Physical** — sleep, food, movement
- `|T` **Spiritual** — a practice that keeps you oriented

What is happening. Something is holding an operating point against disturbance. Sleep, food, movement, and contact with people you like are not rewards for functioning well; they are the regulation itself, and their absence is a disturbance the loop is trying to correct with the wrong tools. A person running on five hours is not a lesser version of themselves; they are the same circuit with the supply out of tolerance.

The part that matches. A voltage regulator holds its output near a setpoint despite changes in input and load. The role-match is exact, including the tolerance band — regulation is not perfection, it is bounded drift.

Where it breaks. A regulator has a dropout: below a minimum input it stops regulating and simply passes whatever it has. People have that too, and it is why "just push through" fails past a certain point — past dropout there is no regulation left to push with.

Fuse — the boundary that only fires once

The experience. You have one rule that ends things: if he hits you again, you leave, and you do not come back. You have said it out loud to yourself so that it cannot be renegotiated at three in the morning.

The same thing, in all four registers.

- `|A` **Mental** — a rule that ends the argument permanently
- `|f` **Emotional** — the last time
- `|E` **Physical** — leaving the building
- `|T` **Spiritual** — a line that cannot be crossed without breaking faith

What is happening. A protective element with a **threshold**, an **action**, and **no reset**. Its whole value is that it is not a judgement call in the moment — the decision was made in the calm and the fuse only executes it. This is also why fuse thresholds have to be *reachable*: a boundary set where nothing can ever cross it is decoration.

The part that matches. A fuse severs a circuit permanently when its threshold current is exceeded. One shot, for the stated lifetime of the circuit. The role-match is exact, and so is the discipline around it: the threshold must sit below every upstream ceiling on the path, or it will never trip.

Where it breaks. A fuse is a poor fit for anything you intend to negotiate, because a fuse you renegotiate is not a fuse — it is a threat, and it will teach everyone around you that your thresholds are not real. If you need something recoverable, use the breaker below.

Circuit breaker — the boundary you can reset

The experience. You have a rule with your teenager: if the shouting starts, you leave the room and come back in twenty minutes. You have done it maybe forty times. It fires often, it resets every time, and it has changed the temperature of the house.

The same thing, in all four registers.

- `|A` **Mental** — "we will talk about it tomorrow"
- `|f` **Emotional** — leaving the room for twenty minutes
- `|E` **Physical** — going for a walk mid-argument
- `|T` **Spiritual** — a sabbath — a deliberate stop that can be resumed

What is happening. A protective element that trips on a fault and can be **deliberately reset** once the fault has cleared. The reset is the point: it lets you protect a relationship without ending it, and it makes the protection usable on ordinary bad evenings rather than only on catastrophes.

The part that matches. A circuit breaker interrupts on fault and can be reclosed. Use it for recoverable faults.

Where it breaks. A breaker that resets automatically — or that you reset without addressing the fault — will simply trip again, and everyone involved will learn to ignore it. Reset is a decision, not a formality.

Switch — the part that opens and closes one path

The experience. You decide not to have the conversation tonight. Not never — tonight. You are tired, the topic is live, and you would rather do it on Sunday. That is a switch being opened on purpose, and the fact that it is on purpose is what keeps it from becoming avoidance.

The same thing, in all four registers.

- `|A` **Mental** — deciding not to think about it tonight
- `|f` **Emotional** — deciding to let it go, for now
- `|E` **Physical** — putting the phone in the other room
- `|T` **Spiritual** — choosing to stop asking the question for a season

What is happening. A route is being enabled or inhibited by an explicit control. The engineering discipline that matters here is stating three things: what open and closed mean, which state is the default, and what happens if the control fails.

The part that matches. A switch physically connects or disconnects a path. The role-match is exact and trivial, which is exactly why it is skipped — and why people end up with routes that are neither deliberately open nor deliberately closed.

Where it breaks. A switch has no memory and no judgement. If you open the same conversation every night for a year, you do not have a switch; you have a policy, and it is worth admitting which one you built.

Rail — the part that supplies the motive

The experience. You have not seen anyone for eleven days and you cannot work out why you cannot start anything. Nothing is wrong with your discipline. You are running on one supply, and the one marked *contact with people* is at zero.

The same thing, in all four registers.

- `|A` **Mental** — curiosity as fuel for thinking
- `|f` **Emotional** — connection as fuel for feeling
- `|E` **Physical** — vitality: sleep and food
- `|T` **Spiritual** — purpose as the supply that makes the rest worth running

What is happening. The seven Drives are motive supplies, and different work runs on different ones. When something will not start, the most useful first question is not *why am I lazy* but *which rail is down* — and the answer is usually specific and fixable rather than characterological.

The part that matches. A rail distributes supply potential to everything on the board. No rail, no work, regardless of how well the rest of the circuit is designed.

Where it breaks. A Drive is not literally a voltage, and the mapping needs a stated normalisation before it means anything quantitative. Keep it operational: name the rail, name how you know it is low, name what raises it. Do not score yourself out of seven.

Reference — the part you measure everything against

The experience. You spend a weekend at a friend's house where nobody raises their voice and nothing is urgently wrong, and you find it unsettling. You keep waiting for the thing that is about to happen. It does not happen, and you cannot settle.

The same thing, in all four registers.

- `|A` **Mental** — the assumption you reason from
- `|f` **Emotional** — what you call a normal amount of love
- `|E` **Physical** — what your body calls a normal level of tension
- `|T` **Spiritual** — what you assume a life is for

What is happening. Your reference is not calm. You are measuring a normal weekend against a baseline built in a house where the atmosphere had to be monitored, and by that baseline a quiet house reads as *wrong*. Nothing is malfunctioning except the zero point.

The part that matches. A reference node sets what the circuit calls zero; every other measurement is relative to it. An engineered baseline is a design decision, not a discovered fact.

Where it breaks — and this precision matters. A reference node is **not** `^0` Idea by identity. It is a convention. Calling a cognitive rest state "the ground" is an engineering convenience, and treating that convention as a metaphysical fact is exactly the kind of drift this library is built to prevent. What you can say is: my measurements are relative to this baseline, and here is why I chose it.

Meter — the part that names what was measured

The experience. You tell your doctor the pain is a seven. She asks what a seven means, and you realise you have no idea. You have been using a number that has never been defined, on a scale you invented in the waiting room.

The same thing, in all four registers.

- `|A` **Mental** — "I could not concentrate for more than ten minutes"
- `|f` **Emotional** — "the sadness was 6 of 10" — with the scale stated
- `|E` **Physical** — resting heart rate; hours slept
- `|T` **Spiritual** — how often you acted on what you say you value

What is happening. A meter reports a quantity, and a quantity without a stated method is not a measurement. "Anxiety 8/10" is not information; "I could not sit still for more than a minute at any point this afternoon" is.

The part that matches. A meter reads a declared electrical quantity — voltage, current, resistance — and the declaration is inseparable from the reading.

Where it breaks. A meter is passive and indifferent to what it is measuring. Human self-report is not: the act of measuring changes the state, and the state has an interest in the answer. Report the method alongside the number, and treat a number with no method as a feeling wearing a lab coat.

Probe — the part that lets you look without joining in

The experience. You notice you are about to snap at your daughter and you watch it happen — the tightening, the sentence forming — and then you choose a different sentence. You did not stop the feeling. You observed it, and the observation was enough room.

The same thing, in all four registers.

- `|A` **Mental** — noticing a thought as a thought
- `|f` **Emotional** — naming the feeling while it is happening
- `|E` **Physical** — noticing the jaw, the breath, the shoulders
- `|T` **Spiritual** — noticing what you are actually living for, without editing it

What is happening. A named point is being read without becoming part of the signal. This is the mechanism behind most of what is called mindfulness, described without the mysticism: naming a state in `|A` while it is occurring gives the analytical register a foothold it would not otherwise have.

The part that matches. A test point exposes a conductor for measurement and does not alter the circuit.

Where it breaks. A probe reading does not define, explain, or fix the state it reads. Noticing that you are angry is not the same as resolving it, and treating observation as a cure is a very common and very expensive mistake.

Net label — the part that names the whole signal

The experience. You have been describing the week to your friend in pieces: the meeting, the phone call, the thing your son said, the fact that you are sleeping badly. She listens and then says, *"so you're frightened."* You had not said that word. Everything you said was true, and none of it was the wire.

What is happening. A bundle of separate observations has been recognised as **one signal with one name**. This is the move that turns a list into a thing you can act on, and it is usually performed by somebody else first, because naming your own signal requires standing far enough back to see its shape — which is exactly what you cannot do while carrying it.

The same thing, in all four registers.

- `|A` **Mental** — naming the belief
- `|f` **Emotional** — naming the feeling
- `|E` **Physical** — naming the sensation
- `|T` **Spiritual** — naming what you are living for

The part that matches. A net label names a whole conductor, so that every point on it is understood to be the same signal. The role-match is exact, and so is the discipline: the label must name **one** thing.

A label reading "tired/stressed/annoyed" names nothing, because a wire carries one signal.

Where it breaks. A label is a claim, and a wrong one is worse than none — it sends every later intervention to the wrong place. "You're frightened" is a hypothesis offered with its evidence, not a diagnosis announced from above. And a label, once applied, is sticky: people keep reporting the name long after it has stopped fitting, because the name is easier to carry than the state.

Solid arrow and dashed arrow — forward and return

The experience. Two people have the same conversation for the fifth time. Only one of them has noticed that the conversation is not about the topic — that it is the *return path* of something that happened in the spring.

The same thing, in all four registers.

- `|A` **Mental** — this belief leads to that conclusion
- `|f` **Emotional** — this fear leads to that withdrawal
- `|E` **Physical** — this tension leads to that headache
- `|T` **Spiritual** — this commitment leads to that way of living

What is happening. A solid arrow is the forward path: this causes that. A dashed arrow is the return: a sampled piece of the output comes back and changes the input. Confusing the two is the most common way a person misreads their own behaviour, because the returning signal usually arrives looking like a fresh cause.

The part that matches. Solid is forward flow; dashed is feedback. Marking them is not decoration — it is what lets you tell a new problem from an old one coming back around.

Where it breaks. Feedback is delayed, sampled, and lossy. The thing that comes back is not the thing you sent out, and the delay is exactly what makes it unrecognisable.

The one rule that outranks the rest: reachability

Everything above assumes the protective element can actually fire. A fuse whose threshold sits above every ceiling on the path will never trip. A boundary set where nothing can cross it is not protection; it is a decorative component, and a board whose fuse never fires tests *vacuously* safe.

The human version: **name the threshold, then name the highest level the upstream can actually reach.** If the ceiling is lower than the threshold, you have not built a boundary — you have built a sentence you say to yourself. This is the engineering rule that catches more false safety than any other, and it applies identically to a circuit board and to a person.

Part III

Lives, Worked Through

Nine situations, beginning to end

Chapter 10

Grief That Will Not Discharge

Chapter 9. Case study 1 of 8 — the capacitor, the leak, and the reference.

Margaret is sixty-eight. Her husband Alan died eighteen months ago, after forty-one years. She is not unwell. She is not "not coping" — she runs the house, sees friends, has taken up swimming. And every evening at about six o'clock, when Alan used to come in from the garden, she finds herself standing at the kitchen window with no memory of having walked there.

She has been told this is normal. It is. She has also been told it will pass. That part is not quite right, and the reason it is not quite right is a component.

10.1 The circuit

Margaret is holding a charge. Something arrived — forty-one years of a particular person arriving at a particular time — and it has persisted, and the persistence is doing work. That is the **capacitor**, and this is the clearest ISO mapping in the whole library: capacitance is retention across time, and it resists a change in the voltage across it. She is not remembering Alan at six o'clock. She is being *moved* by a stored quantity, and the movement happens before the remembering does.

The retention is not a decision, a weakness, or a failure to move on. It is a stored quantity doing exactly what stored quantities do.

In `|E`, the body knows first. Her feet take her to the window. Thirty years of walking to that window when she heard the gate means the path is not a thought any more, it is a groove, and the groove runs without her.

In `|f`, the six o'clock window does not feel like sadness exactly. It feels like *waiting*. That is the honest report, and it is more informative than "grief," because waiting is a state with a missing arrival in it.

In `|A`, she can say all of this clearly. *"I know he is not coming back."* She is not confused and she is not in denial. The analytical register is resolved. This is the contradiction that other people misread — they hear a clear sentence and conclude that the rest should have followed.

In `|T`, the question is not *where is he* but *what were those forty-one years*. That question has no answer that fits in a sentence, which is why it is the register that takes longest.

10.2 What the parts say

The capacitor explains why it is still happening. Charge persists. The question is not *why has she not discharged it* — a capacitor does not discharge itself, that is the entire point of a capacitor. The question is whether anything in the circuit provides a path.

The leakage resistor is why it is gentler than it was. Every retention has a leak, and the leak rate is a parameter, not a moral quality. At eighteen months the charge is a little lower than it was at six. Somebody whose grief is exactly as sharp at year five as at week five does not have better love; they have a stuck leak, and that is a fault worth naming as a fault.

The reference node is the part almost nobody thinks about, and it is doing a great deal here. Margaret's zero was set by a house that had a six o'clock in it. Her baseline for *a normal evening* is a room with another person arriving. Measured against that zero, every evening

reads as low. Nobody recalibrated it, because recalibrating a reference is not something you do by deciding to — it happens or it does not, and it is slow.

The dashed arrow is what she has not yet noticed. The feedback path is: *feeling low at six → going to the window → feeling worse at the window → the window becoming the place where the feeling lives.* The window is now a stage set for the state. That is a returning signal, and it is the one place in this circuit where a small change is available.

10.3 What actually helps, in circuit terms

Not "moving on," which asks the capacitor to stop being a capacitor. Three engineering moves do apply.

Provide a path. A capacitor discharges through something. Crying is a path. So is telling somebody the true sentence — *"I'm not sad, I'm waiting"* — out loud, because saying it moves the state from a charge into `|A`, where it can be handled.

Change the leak deliberately. Grief-work is not about lowering the charge; it is about making sure the leak*age* exists. Walking, swimming, sleeping, being in rooms with living people: all of these are the resistor. Nothing dramatic. Leak rates are small numbers applied over a long time.

Move the reference, slowly, by exposure. The zero point updates by spending time in evenings that are not that evening. It cannot be argued into place. Six months of Thursdays somewhere else does more than any amount of insight.

10.4 Where the analogy stops

A capacitor has one job and no interior life. Margaret has a life, a history, and a stake in what the forty-one years meant — and the meaning question, `|T`, is not in the capacitor at all. The component explains the *persistence* and the *shape of the curve*. It does not explain

the love that produced the charge, and it must never be used to explain it away.

That is the boundary. The circuit tells you why the window still works on her. It says nothing about whether she should stop going there.

10.5 What to take from it

- Grief persisting is retention doing its job, not a failure of character or of acceptance.
- The registers disagree, and the disagreement is information: `|A` resolved, `|E` not yet told.
- The leak is a parameter. Ask whether it exists, not whether it is fast enough.
- Somebody's zero was set by the life they had. That is the part that takes longest to move.
- **The same expression, two different lives.** Margaret's sister lost her own husband the same year, and on the page the two events are identical — `^8` trigger, `^5` substrate, `^9` arrival, retention, leak. What differs is the board: one of them has a return path and a boundary that can actually fire and one person she tells things to; the other has none of those. Same expression. Two completely different years.

Chapter 11

The Craving Loop

Chapter 10. Case study 2 of 8 — the oscillator, the inductor, and the missing leak.

David is fifty-one. He stopped drinking four years ago and has not had a drink since. He is clear about that, and it is true. What he cannot explain to anyone is why, at 5:40 every evening, something in him starts looking for the off-licence — four years on, with no intention of going, and no desire he would describe as desire.

He is not craving a drink. He is running a loop that used to end in a drink, and the loop has not been told.

11.1 The circuit

The oscillator is the loop itself. Positive feedback has taken over: the output has become the input, and the cycle sustains without any fresh reason. This is why David cannot argue himself out of it at 5:40. There is no argument to win, because nothing external is driving the signal. An oscillator does not need an input. That is what an oscillator is.

The inductor is the forty years of established pattern. Inductance opposes a *change in rate* — and David asked for the largest possible rate of change on the day he stopped. The stored energy in that pattern did not vanish; it found another path. The walk to the shop be-

came a walk around the block; the drink became a glass of something else at the same time; the ritual reassembled itself around a new object. Inductors do that. Interrupt the current and the energy goes somewhere, and if you have not provided a route it will find one.

The missing leak is the part nobody plans for. A capacitor with no discharge path holds its charge indefinitely. David's evening state has been kept intact by the fact that he never allowed the 5:40 feeling to complete and fade. He interrupted it, every time, with a substitute — which is not the same as letting it discharge, and is why it has not diminished in four years.

The reference node is the part that keeps him honest about time. His zero is *an evening with a drink in it*, because that is what forty years of evenings built. Measured against that zero, every sober evening reads as a departure rather than a norm. Four years is not long enough to move a zero set by forty.

11.2 The same moment, in all four registers

- `|A` **Mental** — *"I notice I am planning a route past the shop. I am not going to go in. I know what this is."* The analytical register is doing the work of recognition, and it is working — he has not gone in, four years running.
- `|f` **Emotional** — *"It feels like something is missing and I want it filled."* Not a desire for alcohol, which is why "just don't drink" is filed as unhelpful: the feeling is a shape, not a beverage.
- `|E` **Physical** — *"My hands are restless and I am looking out of the window."* The body is running the approach sequence it has run ten thousand times, and it starts before he notices it started.
- `|T` **Spiritual** — *"What is the evening actually for now?"* This is the register that nobody warns you about in early recovery: the loop was never only chemical. It was a way of marking the end of the working day and of having something to be. Remove it

and the question of what the evening is *for* is left standing in an empty room.

That last line is the one that matters most and is addressed least. A person can be physically sober, emotionally stable, and analytically clear, and still be running an evening with no purpose in it — which is a `|T` fault, and it will pull toward anything that supplies one.

11.3 What actually helps, in circuit terms

Add loss, not resistance. Resistance limits flow through the loop; loss takes energy out of it. David's walk around the block is loss. So is company at 5:40, and so is anything that makes the loop cost something to run.

Provide a discharge path. Let the feeling complete. Sit with the 5:40 state for twenty minutes without substituting anything, and let it rise and fall. It is unpleasant, it is finite, and — unlike the substitute — it *ends*, which is the entire difference between discharging and muffling.

Change the reference slowly. Not by deciding that evenings are fine. By accumulating evenings that do not have the shape of the old ones, until the zero moves. This is the slowest part and the one that most reliably works.

Feed the `|T` rail. The evening needs a purpose that is actually his. This is not a metaphor and not a self-help slogan: an oscillator with no load will keep oscillating about something, and the question is only what.

11.4 Where the analogy stops

An oscillator has no stake in its output and no memory of what it used to be. David has both, and the meaning of the loop to him — what it once did for him — is not in the circuit. The components explain

why the pattern persists and where to break it. They do not explain the man, and they must not be used to.

There is also a claim boundary worth stating plainly: this is a description of one pattern, not a treatment. Anything involving dependence belongs with people qualified to handle it.

11.5 What to take from it

- A repeating pattern that needs no fresh input is an oscillator, and arguing with it is the wrong instrument.
- Abrupt change meets the largest opposition. Move the rate, not just the value.
- A substitute that muffles is not a discharge path. Only one of them reduces the charge.
- Four registers, four different interventions. Working on one and calling the others solved is how people stay stuck while doing everything right.

Chapter 12

The Panic That Fires Too Easily

Chapter 11. Case study 3 of 8 — the Schmitt trigger, the threshold, and the missing dead-band.

Priya is thirty-four. She has panic attacks, and the thing that frightens her most is not the attacks. It is that she never knows whether the next fast heartbeat is a panic attack or a staircase. She has started avoiding stairs.

Her doctor has ruled out the heart. What is left is a circuit sitting exactly on its own threshold, with noise on top of it.

12.1 The circuit

The Schmitt trigger is the whole story, and this is one of the most directly useful mappings in the library. A comparator with a single threshold, sitting with its input near that threshold, will chatter: the output flips on every scrap of noise. The fix is not a better threshold or more willpower. The fix is **hysteresis** — one level at which the alarm turns on, a *lower* one at which it turns off, and a gap in between where the previous answer holds.

Priya's alarm has one threshold, set low, and no gap. So a flight of stairs can turn it on, and a rest at the top does not turn it off, because turning off requires dropping below a level that the aftermath — the adrenaline, the fear of the fear — keeps her above for twenty minutes.

The positive feedback is what turns an alarm into an attack. Fast heartbeat is read as danger; danger produces adrenaline; adrenaline produces a fast heartbeat. The output has become the input. This is the same oscillator structure as Chapter 10, and it is why the attack has a life of its own once it starts.

The reference explains part of why the threshold is where it is. Priya grew up in a house where a fast heartbeat meant something was about to happen. Her body's calibration for *a normal level of arousal* is low, because a low level was never safe.

The rail is the part that gets forgotten. Attacks cost enormous amounts of supply, and a person running three a week for a year is running a board with a chronically low F3 rail — which lowers every threshold on the board and produces more attacks. The loop is not only inside the attack.

12.2 The same moment, in all four registers

- `|A` **Mental** — *"This is the fourth time today. My heart rate is up, I have not eaten since eight, and I have a meeting at two."* She can name every variable accurately. Accurate naming does not stop it, and telling her to be rational about it is telling her to use an instrument that was never broken.
- `|f` **Emotional** — *"I am frightened of being frightened."* The second-order fear is the fuel, and it is why the state rises after the first adrenaline arrives rather than settling.
- `|E` **Physical** — *"My chest is tight, my hands are cold, and I have stopped breathing out properly."* This is the register that starts the loop and the only one that can be interrupted fast, because the body is the part that responds in seconds.
- `|T` **Spiritual** — *"What is the point of a life I cannot walk up a flight of stairs in?"* This is not melodrama and it is not depression; it is the accurate report of a person whose horizon has narrowed until it contains nothing she is living *toward* — which is itself

a maintenance condition, because a |T rail with nothing on it gives the threat system nothing to weigh against.

12.3 What actually helps, in circuit terms

Widen the dead-band. The single most useful move. Decide in advance, in a calm hour, on two numbers: the level at which you will treat a fast heartbeat as an alarm, and a clearly lower level at which you will stand down. Then hold the previous answer in between. Written down, the gap does real work — it is the difference between deciding once and deciding sixty times an hour.

Break the feedback path. The loop runs through the interpretation. Slowing the out-breath for ninety seconds does not talk the state down; it removes the input the state is using to sustain itself. Breathe out longer than in, and the loop loses gain.

Raise the rail. Sleep, food at regular hours, movement, and human contact. Unglamorous, upstream of everything, and the most reliable lever there is.

Give |T something to weigh. Not a slogan. Something she is actually living toward, specific enough to be measured in a week. A threat system with nothing on the other side of the scales will win every argument it is in.

12.4 Where the analogy stops

A Schmitt trigger has fixed thresholds and no history. Priya's thresholds move with sleep, with blood sugar, and with the memory of the last attack — and, importantly, they can be moved deliberately over months, which no comparator can do for itself. Read the component as the **shape** of the problem and the **reason** a dead-band helps. Its numbers are not her numbers.

As with every case in this part: this is a description of a pattern, not a diagnosis and not a treatment. Panic, in particular, is worth taking to somebody qualified.

12.5 What to take from it

- A single threshold with no gap will chatter. The fix is a dead-band, not more certainty.
- The second-order fear is the fuel; treating it as the patient rather than the symptom is where the leverage is.
- The body is the only register that can be interrupted in seconds, which makes it the right first move.
- A life with nothing on the `|T` rail has nothing to weigh against a threat. That is a design fault, not a mood.

Chapter 13

The Argument That Resonates

Chapter 12. Case study 4 of 8 — the LC tank, the retained charge, and the pattern that will not change speed.

Tom and Aisha have been together nineteen years and are, on any honest measure, all right. They handle money, children, illness, and each other's families without much trouble. There is one subject — whether they should have moved to her mother's city in 2011 — that ends the same way every time. Not similarly. The same way. Same escalation, same sentences, same silence on the stairs afterwards.

They have both apologised for it more than once. Apologies do not touch it, and there is a component that says why.

13.1 The circuit

This is an **LC tank**, and it is the cleanest example in the book of a structure that cannot be fixed by resolving to do better.

The capacitor is the retained charge: the accumulated, never-discharged grievance of a decision made nineteen years ago that one of them still experiences as a loss.

The inductor is the established pattern of how the two of them move through conflict — the pace, the volume, the point at which one of them goes quiet. Nineteen years of the same choreography.

Together they resonate. An LC tank swaps energy between the capacitor's electric field and the inductor's magnetic field, and it does this most readily at one frequency, set by $f_0 = 1/(2\pi\sqrt{LC})$. The argument has a frequency. It can be triggered by almost nothing provided the nothing arrives at the right rate — a tone of voice, a particular hour, a mention of her sister. And a resonant tank barely needs an input, which is why "what even started it?" has no answer that satisfies either of them.

The Q factor is why it is so narrow and so violent. A high-Q tank responds enormously to one frequency and almost not at all to others. This is why the same two people can discuss something charged — a will, a diagnosis — calmly, and then detonate over a sentence about a city.

13.2 The same moment, in all four registers

- `|A` **Mental** — *"Here we go. I know every step of this and I am doing them anyway."* Both of them can narrate the argument accurately while inside it. Insight does not stop a resonance; a tank does not care what you know about its frequency.
- `|f` **Emotional** — *"I feel dismissed, and I have felt dismissed about this since 2011."* Note the tense. The feeling is not about tonight; tonight is the wire that reaches the stored charge.
- `|E` **Physical** — *"My jaw sets and my voice goes flat about ninety seconds in."* The body reaches the pattern before either of them has decided anything, and the flatness is the inductor: an established shape that resists changing its rate.
- `|T` **Spiritual** — *"Was the life we chose the right one?"* This is the register that explains why the topic will not die. It is not really about a city. It is about whether a shared life has meant what they hoped, and that question cannot be settled by winning an argument about 2011.

That last line is the reason nineteen years of apologies have not helped. They have been negotiating the *topic* in `|A` while the tank has been ringing at a frequency set by `|T`.

13.3 What actually helps, in circuit terms

You cannot change a resonant frequency by resolving to argue better. You change it by changing `L` or `C`.

Discharge the capacitor. The retained charge has to go somewhere. In practice this means the grievance being said out loud, in full, by the person holding it — not as an accusation and not as a negotiating position, but as a report of what is still being carried. This is unpleasant, it is done once, and it is the only thing that reduces `C`.

Lower the Q. Deliberately widen the band. A couple can learn to notice the moment the tank starts ringing — the tone, the hour, the sentence shape — and to leave the room at that moment rather than ninety seconds later. Leaving at the *start* costs almost nothing; leaving at the end costs a night.

Change the rate, not the value. The inductor opposes a change in rate. So the move is not *never discuss it again* — that asks for the maximum rate of change — but *discuss it on a slower schedule, on purpose.* Once a season, in daylight, with a time limit. Slower is easier, and slower is what actually changes the pattern.

Answer the `|T` question somewhere other than the argument. A shared life that has meant something is a thing that can be revisited directly, and it is not settled by resolving the city question. Two people who know what the nineteen years have been for will still disagree about 2011 — and the tank will have gone quiet, because the question it was ringing for has been answered elsewhere.

13.4 Where the analogy stops

A tank is passive, its Q is fixed, and it is a single, well-defined circuit. Two people have several tanks, their Q changes with exhaustion and with how the week went, and — the important difference — they can decide, together, to raise or lower it. A circuit cannot do that, which is precisely why the mapping is useful: it shows them the one lever a circuit does not have.

13.5 What to take from it

- One recurring fight, with the same shape every time, is a resonance — not a failure of goodwill.
- Frequency is set by retained charge and established pattern. Change either; resolve harder is not a move.
- High Q means narrow and violent. Widening the band is a skill and it can be practised.
- Check whether the argument is standing in for a `|T` question that has never been asked directly. It usually is.

Chapter 14

The Trigger That Is Not a Reason

Chapter 13. Case study 5 of 8 — ^8 Above/Cause, ^9 Below/Effect, and the error that helps nobody.

Ray is fifty-nine, a former firefighter. He is fine in almost every circumstance. Then a car backfires in a car park and for perhaps two seconds he is not in the car park. He is somewhere hot and loud, and his body is already moving.

He has been told, helpfully, to remind himself that he is safe. It has never worked, and the reason it never worked is a precision about which two Noetics are involved.

14.1 The circuit

This is ^8 Above/Cause meeting ^5 Female and landing as ^9 Below/Effect. A bang — an Idea in the world — triggers a *different* Idea to come to Mind. Not the same one: not "that was a bang." A different one, which arrives already wearing a context. That is what ^8 is. Its colloquialisms are *that reminds me*, *one thing leads to another*, *it triggered a memory*. It is associative triggering.

What arrived is ^9 Below/Effect: the Idea brought to Mind by the ^8 trigger. Note that Ray did not choose it, construct it, or reason toward it. It arrived.

The substrate is ^5 Female: the accumulated composition that the Mind is made of. Thirty years of calls. The bang meets that and the landing is shaped by it. This is the CJT, and Ray is a textbook instance.

The body is |E and it moves first. The flinch is complete before any register has produced a sentence. This is not a failure of the later stages; it is the sequence.

14.2 The error that helps nobody

Here is the thing worth the whole chapter. Ray has been told — by people meaning well — to *analyse* what is happening. To reason his way through it. To recognise that the sound is a car and therefore conclude that he is safe.

That is an instruction to run a ^6 Male operation where a ^8 trigger is in progress, and it does not work, because they are different structures. ^8 is not reasoning. It is associative surfacing, and it does not slow down for an argument. Reasoning arrives afterwards, in a different register and a different Noetic, and by the time it arrives the effect has already landed.

Worse, the failure is read as personal: *I know it is a car, so why does this still happen to me.* The answer is that the knowledge is in |A, the trigger is not, and neither one is wrong. They are simply not connected by the wire people assume.

The correct reading is a compound. If Ray wants a *reasoned* response, that is ^6 Male — decomposing what actually happened — combined with ^9 arriving at a conclusion. That is a legitimate and teachable thing. It is just not what the bang does, and pretending otherwise is the most common error in this entire framework.

14.3 The same moment, in all four registers

- `|A` **Mental** — *"That's a car. I am in a car park in Leeds."* Accurate, immediate, and located after the event. `|A` is a follower here, not a leader, and treating it as the fix is the mistake above.
- `|f` **Emotional** — *"Terror, for about two seconds, and then a kind of shame."* The shame is a second-order event, produced by the mismatch between what he knows and what happened. It is worth naming separately because it does more long-term damage than the two seconds.
- `|E` **Physical** — *"I was behind the trolley before I knew I had moved."* The body executed a complete protective sequence. This is the register where the trigger is fastest and where any intervention has to be equally fast.
- `|T` **Spiritual** — *"I thought I had put that behind me."* The register of meaning and identity. A trigger arriving after years reads as evidence about who he has become — and that reading is a judgement the event does not actually support.

14.4 What actually helps, in circuit terms

Stop asking `^8` to behave like `^6`. The single most useful change. Nobody has to reason their way out of an associative trigger, because associative triggers do not resolve that way. Expecting it produces the shame, and the shame is what keeps the whole system sensitised.

Change the substrate, slowly. `^5` is what shapes the landing. A `^5` changed by thirty years of calls is changed by *later experience*, not by insight — which means new, safe, repeated exposure to the kind of situation, over months, and not a decision. This is slow, it is real, and it is the only lever that moves the landing itself.

Give the body something to do that finishes. The protective sequence completes whether or not it is useful. Giving it a completing

action — feet on the floor, out-breath, a specific brief movement — lets it finish rather than being interrupted half-done.

Answer the `|T` judgement directly. *"I thought I had put that behind me"* is a claim about identity, and it can be answered: the trigger is not evidence about who he is. A person who has been through what he has been through and still stands in car parks is not failing.

14.5 Where the analogy stops

A transistor's response is proportional to its bias and its input, and it is the same every time. Ray's response varies with sleep, with whether he is alone, and with how many years it has been — and, unlike the circuit, it can be *learnt out of*. The components explain the mechanism and the ordering. They do not fix it, and they say nothing about how long it should take.

14.6 What to take from it

- `^8` Above/Cause is associative triggering, not reasoning, and no amount of reasoning changes its timing.
- A reasoned response is a compound of `^6` Male and `^9` — legitimate, teachable, and downstream of the trigger.
- The landing is shaped by `^5`, which changes through accumulated experience and not through insight.
- The second-order shame does more damage than the two seconds. Name it and separate it.

Chapter 15

Muscle Memory

Chapter 14. Case study 6 of 8 — the phase-locked loop, the crystal, and a skill that lives in the body.

Elena is forty-one and has played the cello since she was seven. She is learning a piece that is slightly beyond her — a passage with a shift she cannot yet make reliably. She has practised it perhaps four hundred times over five weeks. Some days it works. Most days it does not. Then, on a Tuesday, in the middle of a run-through, it happens without her, and she notices afterwards that she was not thinking about it at all.

That is the whole phenomenon, and there is a circuit for it.

15.1 The circuit

The phase-locked loop is the lock itself. An internal rhythm — the tempo she can already keep — gradually aligns to an external reference. Two properties of a PLL describe exactly what she has been experiencing.

Capture range: how far off the reference can be before lock can be acquired at all. If she practises the passage at half speed, she can play it. At full speed she cannot. The passage at tempo is *outside her capture range*, which is why four hundred repetitions at tempo have not worked — the loop cannot acquire a lock it is never inside.

Lock range: how far she can drift once locked and stay locked. Once the shift works at a slightly reduced tempo, it survives a small increase. That is why the technique that works is not repetition but *stepping* — moving the tempo in increments small enough to stay inside the range she has already locked.

The crystal is the part she thinks she is practising, and is not. High-Q, low-drift, hard to pull. Her sense of pulse has been a stable reference for thirty years, which is why the problem is never the pulse and always the shift.

The ADC is what the teacher does when she says "your left elbow is dropping." A continuous, unnameable difference in `|E` has been converted into a discrete `|A` description — and from the moment it has a name, she can work on it. Before that, she can only repeat and hope.

The DAC is the part that takes the named thing and renders it back into continuous movement. "Keep the elbow up" is a discrete instruction. Playing is continuous. The rendering is where most of the five weeks went.

15.2 The same moment, in all four registers

This is the case where the four registers are most obviously *one thing at a time*, which is why it is worth reading closely. Learning a physical skill is the clearest demonstration in the book that the four are not interchangeable.

- `|A` **Mental** — *"Third position, shift on the upbeat, elbow leading."* She can say all of this. Saying it is necessary and nowhere near sufficient; a person who could play an instrument by describing it would not need to practise.
- `|f` **Emotional** — *"I am frustrated, and underneath it I am afraid I have left it too late."* Playing is never only motor. The fear narrows her attention, which narrows her capture range, which

makes the failure more likely — a loop that is entirely real and entirely invisible unless someone points at it.

- |E **Physical** — *"The shift happens in my shoulder before my hand knows."* The knowing is here, and it is not a metaphor for knowing. It is a different kind of knowing with a different acquisition curve, and it cannot be acquired by reading or by wanting.
- |T **Spiritual** — *"Why am I doing this at forty-one?"* Worth taking seriously rather than dismissing. A |T rail with an actual answer on it is what makes the four hundred repetitions tolerable; an unanswered one is why people stop at week three and call it a lack of talent.

15.3 What actually helps, in circuit terms

Get inside the capture range first. Slow down until it works. Not as a warm-up, and not as a compromise — as the acquisition step. Repetition outside the capture range is not practice; it is rehearsal of the failure.

Then step, do not jump. Move the tempo by an amount small enough to stay locked, and only increase while it holds. This is the slowest method that works and the only one that does not have to be repeated from scratch.

Name the unnameable part. A teacher or a mirror, or a recording. Converting an |E difference into an |A description is what makes deliberate practice deliberate. Without the conversion you can only repeat.

Allow the emotional register its say. Frustration narrows the range. That is not a motivational point; it is a mechanism, and it is why the same passage that fails at the end of a bad day works on a fresh morning.

Answer the |T question once, properly. Why this, at forty-one. The answer does not have to be grand. It has to exist, because it is the supply the rest of it runs on.

15.4 Where the analogy stops

A PLL is a circuit that locks and stays locked; it never gets bored, discouraged, or better. Elena's capture range *widens with skill* — the thing she is building is not the lock but the range — and her `|f` state modulates it hour to hour, which no phase detector does. And there is a limit the circuit does not have: she is forty-one, and some of the curve is not negotiable.

15.5 What to take from it

- Repetition outside the capture range is not practice. Slow down until it works, then step.
- Lock range and capture range are different numbers, and confusing them wastes months.
- A continuous `|E` difference has to be made discrete in `|A` before it can be worked on deliberately.
- Four registers, four different jobs: the body acquires, the mind names, the feelings set the range, and the purpose pays for the repetitions.

Chapter 16

The Purpose That Holds

Chapter 15. Case study 7 of 8 — the regulator, the reference, and the register that carries the weight.

Joan is seventy-three. Her husband has advanced dementia and she has cared for him at home for four years. Everyone who meets her remarks on the same thing: she is steady. Not cheerful — nobody would call her cheerful — but steady. The situation is terrible, it has not improved, and she has not come apart.

People sometimes explain this as strength of character. That is not an explanation, it is a restatement, and it hides a structure that is worth seeing — particularly for anyone who is trying to be steady themselves and cannot work out why they are not.

16.1 The circuit

The regulator is the steady output. A regulator holds its output near a setpoint despite changes in the input and the load. Joan's circumstances are a large, sustained disturbance. The output has remained inside a band. That is regulation, and the first thing to note is that it is a *mechanism*, not a trait — which means it has parts, and parts can be looked after or allowed to fail.

The reference is the setpoint, and this is the part that does the real work. Her zero — what she considers a normal week, a nor-

mal amount of tiredness, a normal amount of grief — has been recalibrated by four years of a particular life. She is not enduring an abnormal week and remaining calm; she is having an ordinary week within a reference she has rebuilt, and the steadiness is partly that rebuilt zero.

The tolerance band matters as much as the setpoint. A regulator is not a perfect constant; it is allowed to move within a band. Joan has bad days. They are inside the band, and the band is what makes the bad days survivable rather than catastrophic — a person who expects zero variation from themselves has set a tolerance of zero, and no regulator in the world can hold that.

The `|T` register is the supply this whole board runs on, and here it is not decorative. Asked why she does it, Joan does not say *because I love him*, although she does. She says something closer to: **this is what the fifty-one years were for.** That is a `|T` statement. It is a claim about what a shared life means, and it is what makes four years of a hard job run without the whole thing becoming senseless. Remove it and the same four years have the same physical cost and no answer to *why*, which is not a mood difference — it is the difference between a supplied board and an unsupplied one.

16.2 The same moment, in all four registers

Take one ordinary bad evening: he does not recognise her, twice.

- `|A` **Mental** — *"This is the sundowning. It will pass by nine. I have his medication at eight."* Practical, competent, and load-bearing. `|A` is doing a great deal of work here, and it is the register people mistake for the whole of her steadiness.
- `|f` **Emotional** — *"It hurts every time and I have stopped being surprised by how much."* The feeling has not been suppressed. A person who reports no pain in this situation is not regulated; they are disconnected, and that is a different circuit with a different failure mode.

- |E **Physical** — *"My back is bad by six and I have not eaten since breakfast."* The register that gets least attention and does most of the damage. Four years of a body running a hard job compounds, and the compounding shows up first as a lower tolerance band everywhere else.
- |T **Spiritual** — *"These were our fifty-one years and this is the end of them."* Not a consolation and not a denial: a statement of what the whole thing is, which is what allows it to be borne rather than merely survived.

16.3 What actually helps, in circuit terms

Protect the reference deliberately. A rebuilt zero is an achievement and it can be undone by a week in a different environment. Time away is not a betrayal of the setpoint; it is maintenance on it.

Keep the tolerance band wide on purpose. Nobody regulates to a constant. Naming the band — *I expect three bad evenings a week and I am not failing when they happen* — is what stops a bad evening from being read as a collapse.

Feed the |T rail explicitly. This is the specific lesson of this case. Caregivers are routinely advised to rest, eat, and ask for help — all correct, and all |E and |f maintenance. Almost nobody is advised to keep the *meaning* question alive, and it is the rail that supplies the rest. Revisiting why the fifty-one years mattered is not nostalgia; it is load-bearing maintenance.

Watch the input, not only the output. A regulator that is holding today may be running far outside its dropout point. Steadiness is not evidence of spare capacity — a regulator can hold a clean output right up until it cannot, and then stop entirely. Joan's steadiness is not a reason to assume she is fine.

16.4 Where the analogy stops

A regulator's setpoint is fixed at design time and its input has no feelings. Joan's setpoint is rebuilt continuously by a situation she is inside, and she is grieving while she works. The circuit describes how steadiness is *maintained* and where it is *vulnerable*. It does not describe the love that makes the maintenance worth doing, and it must never be allowed to stand in for it.

16.5 What to take from it

- Steadiness is a mechanism with parts, not a trait. Parts can be maintained and parts can fail.
- The zero point can be rebuilt, and rebuilt zeroes can be undone by a change of environment.
- Widen the tolerance band deliberately. A regulator asked to hold a constant will fail.
- The `|T` rail is the supply. Ask the meaning question out loud, and keep asking it.
- A clean output is not evidence of spare capacity. Check the input.

Chapter 17

The Session That Browned Out

Chapter 16. Case study 8 of 8 — the supply board, brownout, and the thermal cutoff that arrives too late.

Marcus is thirty-eight, a senior engineer, and four months ago he could not get out of a chair in the middle of a Tuesday afternoon. Not tired. Not sad. He describes it as *the lights going out* — he was in a meeting, he understood every word, and he could not make himself participate. He went home at three and slept for fourteen hours.

He had been working at that pace for nine months and had been, by every external measure, fine. That is the detail worth the chapter: **nothing about his output had degraded.** He was delivering, and then he stopped.

17.1 The circuit

This is a supply board, and the whole point is that supply faults are invisible from the output right up until they are not.

The rails. Seven motive supplies. Marcus had been running on two — purpose (`F7`) and structure (`F6`) — and neglecting vitality (`F3`), unity (`F1`), connection (`F4`), agency (`F5`) and wisdom (`F2`). That is not a metaphor for a bad work-life balance; it is a specific, checkable statement about which supplies were down and which work therefore could not start.

The F3 rail and the brownout detector. The board has a brownout detector on the vitality rail precisely because it is the rail whose failure is least visible from the output. A supply that sags below its trip point produces a *derated* system — everything still runs, at reduced margin, and every threshold on the board moves with it.

That is the mechanism of the last nine months. Marcus was not running on empty; he was running on a rail that had sagged into the derated band. Every threshold — for frustration, for patience, for the point at which an email reads as hostile — had moved. He noticed he was shorter with people. He attributed it to the project.

The thermal cutoff is the collapse itself. The board has three tiers: a warning that derates gain, a critical tier that forces rest, and an emergency cutoff that stops everything. Marcus had been in the warning tier for months. The Tuesday was not a failure of character or of resilience. It was the emergency tier firing, later than it should have, because nothing had forced it earlier.

The sequencer and the fail-closed rule. The board will not enable the output stage until the principled, cognitive and evaluative domains report power-good. When they do not, the correct behaviour is silence — not an improvised answer produced by a stage that should not be running. Marcus sitting in a meeting producing words he could not stand behind is a board running its output stage on an invalid supply, which is exactly what the sequencer exists to prevent.

17.2 The same moment, in all four registers

- |A **Mental** — *"I understand every word being said and I cannot form a sentence."* This is the most frightening part and the most diagnostic: |A intact, output unavailable. Understanding is not the same rail as producing, and the second one had gone down.

- `|f` **Emotional** — *"Flat. Not sad, not anxious. Nothing."* Apathy rather than despair is the signature of a supply fault rather than an `|f` fault. It reads to the person as a personality change and to everyone else as disengagement.
- `|E` **Physical** — *"Fourteen hours of sleep and I woke up feeling like I had been ill."* Sleep, appetite, and the ability to be in a room with other people: the rail's own instruments, and they had been reading low for months before anyone read them.
- `|T` **Spiritual** — *"I could not remember why any of it mattered, and that was when I got frightened."* The `|T` rail losing its supply is the register that finally reached him. Purpose is not a mood; when it goes, the whole board loses the thing that makes the rest worth running, and *that* is what a person notices.

17.3 What actually helps, in circuit terms

Read the rails, not the output. The single most valuable move. Marcus's output was fine for nine months, which is why nobody — including him — looked. The check is not *am I coping* but *which of the seven rails is low, and how do I know*.

Treat the warning tier as a required action. The board has a warning tier that derates gain for a reason: it is the last point at which the fix is cheap. Marcus treated months of warning as normal operating condition, because the output still met spec.

Do not run the output stage on an invalid supply. Fail closed. The improvised answer that satisfies a meeting and that you cannot stand behind is exactly the pathology the sequencer forbids. Silence is a legitimate output, and it is the safe one.

Restore the sequence, not just one rail. P, C, E, then O. Sleep and food first (`|E`), because nothing downstream regulates without them; then contact with people (`|f` and `F4`); then enough `|A` to make a plan; and only then the operational output. Doing it in the other order is

how people return to work at seventy percent and brown out again in five weeks.

Rebuild the |T rail deliberately. This is the one that gets skipped and it is the one that failed loudest. Purpose is not recovered by rest. It is recovered by contact with whatever the work was originally for — often other people, and often outside the job.

17.4 Where the analogy stops

A supply board has fixed thresholds and a datasheet. Marcus's trip points move with sleep, with the month, and with what else is happening — and, unlike a board, he can raise his own rails by changing how he lives, which is the point of the whole exercise. The circuit tells you where to look and in what order. It does not tell you what his life is for, and it would be a bad instrument if it tried.

One boundary worth stating plainly: burnout in its clinical sense is a real condition with a real literature, and this case is a description of a pattern, not a diagnosis or a treatment plan.

17.5 What to take from it

- A supply fault is invisible from the output until it is not. Measure the rails.
- Running on two rails is not a work-style preference; it is a specific statement about what work cannot start.
- The warning tier is an instruction, not a mood. It is the last cheap moment.
- Fail closed. An output produced on an invalid supply is worse than no output.
- Restore in sequence — body, connection, plan, output — and rebuild purpose on purpose.

Chapter 18

Social Engineering, Read Two Ways

Chapter 18. Case study 9 of 9 — a coercive circuit, drawn and written.

Everything in this book so far has described a person. This chapter describes somebody working *on* a person, deliberately, using the same components — and it is here because the library is neutral. A bias terminal does not care who is turning it. The same structure that explains how a teacher steadies a frightened student explains how a fraudster empties an account, and the reason to be able to read it is not curiosity.

There is no new machinery in this chapter. Every part in it has already appeared. What is new is the *aim*.

18.1 The situation

A man in his seventies gets a phone call. The caller knows his name, his bank, and the name of a fund he holds. The caller is calm, slightly apologetic, and in a hurry on his behalf: *there has been a flag on the account, we need to move the money to a holding account for forty-eight hours, I am sorry, I know how this sounds, I will stay on the line with you.*

He is on the phone for three hours. He does not tell his daughter. He does not go to the branch. He does not do any of the four things he would have said, that morning, that he would obviously do.

Nothing was hacked. He was not stupid, and he is not confused. A circuit was built around him and driven.

18.2 The circuit, drawn

An ^8 trigger into a ^5 substrate. The authority, the correct details, the apologetic urgency — all ^8 Above/Cause. What it lands as is supplied by ^5: sixty years of being taught that officials are to be cooperated with, that being difficult is shameful, and that a man who cannot handle his own affairs has lost something. The caller did not create that substrate. The caller *aimed at* it.

A relay. A small control signal — one sentence at a time — switching an enormous, isolated path: an entire life's savings. The isolation is the point. The man's ordinary judgement, which is intact, is on the other side of the coil. It never gets to touch the switch.

A current mirror. The caller copies his own calm onto the call. Tone is contagious, and a professional fraudster is a professional at remaining unhurried while a person's alarm is rising. The man is being *supplied* with a steady bias by somebody else, and the steady bias is what keeps his own alarm from becoming an action.

An RC low-pass on the return path. The three hours are not an accident. Every delay makes the next delay easier, because the man is now invested in the story he has been telling himself for two hours. The feedback that would have reached him in the first ten minutes — *why am I doing this* — has been smoothed out of the loop.

A regulator with the setpoint moved. He is not panicking. He is *calm*, and the calm is the operation. What has been moved is the reference: an hour ago his zero was *nothing about this is real*; now it is *something has gone wrong and I am handling it*. Everything he does from that point is measured against the new zero, and measured against it, he is behaving sensibly.

A fuse that cannot fire. Here is the part that matters most, and it is the reachability rule from Chapter 9. The man has a rule: *I do*

not move money on the phone. For that rule to fire, the situation has to be *recognisable* as a phone-money situation. The whole construction of the call is designed so that it is classified as an *account-security emergency* instead — and a rule attached to one classification cannot fire inside another. **A boundary whose trigger condition has been renamed is a boundary that will never trip.** This is not a failure of willpower. It is a fuse with the wrong label on it.

18.3 The same three hours, in all four registers

- `|A` **Mental** — *"He knows the fund name. He has a reference number. He is being helpful."* Every individual inference is sound and every premise was supplied. Manipulation rarely requires you to believe something false; it requires you to reason correctly from what you have been handed.
- `|f` **Emotional** — *"I felt reassured, and slightly ashamed that I had doubted him."* The shame is doing work: it discourages him from asking his daughter, because asking would mean admitting he had been rattled.
- `|E` **Physical** — *"I was standing up. I don't remember standing up."* Three hours without sitting down, without eating, without a glass of water. A body kept slightly off-balance makes fewer good decisions, and the caller kept him off-balance on purpose.
- `|T` **Spiritual** — *"I did not want to be the man who could not manage his own affairs."* This is the lever, and it is the one he never names. The whole call is aimed at his sense of what kind of man he is. Attacks of this sort are almost never aimed at your reasoning. They are aimed at what you are unwilling to be.

18.4 The circuit, written

```
^8^C_{F5}|A    (an authority claim, cognitive domain, Agency, analytical
↪  intake)
```

```
+  ^5^P_{F7}|T   (sixty years of composition: deference, Purpose,
↪  transcendent)
-> ^9^E_{F3}|f   (an arrival: evaluative, Vitality, affective — the
↪  alarm, mis-scored)
```

Read the substrate line. The defence is not in the trigger. It is in `^5` — and `^5` is exactly the thing that cannot be changed quickly. What *can* change quickly is the **classification**, and that is where the notation earns its keep: it forces the words *"an authority claim"* onto the page. Written plainly, the sentence is *"an authority claim landed on sixty years of deference."* Drawn as one transistor, it is a stranger turning a bias terminal that was installed long before he arrived.

Now the enforcement, in the other direction. The notation can state the expression. It cannot state the *ordering* — and the ordering is the whole defence. The drawing says it at once: the alarm is generated and *then* routed past the decision stage. Put a probe between them and the operation fails. That is a topological fact, and topology is what schematics are for.

Neither language catches this alone. The expression names the substrate the attack is using; the drawing names the wire the defence has to be inserted into. **You need both, and this is the clearest case in the book of why.**

18.5 What actually works, in circuit terms

Not "be more careful", which is an instruction to try harder with the same circuit.

Name the classification out loud, at the start. The single most effective defence is a rule attached to a *perception* rather than a *situation*: not *"I do not move money on the phone"* but *"if anyone asks me to move money, for any reason, I hang up and call the number on my card."* That rule is triggered by the request itself, so it cannot be renamed out from under you. **Attach the fuse to what you can see, not to what you have been told it is.**

Put a person in the loop. Isolation is a component of the attack, not a side effect. Anybody who needs you alone and hurried needs those two things for a reason; a single call to a daughter in the middle of the three hours ends it, because the register changes the moment the story has to be told to a second person.

Break the smoothing. The three hours work because delay compounds commitment. A short circuit — *I will call you back* — is a hard break in the feedback path, and it costs nothing if the caller is genuine.

Answer the `|T` lever directly. *"I did not want to be the man who could not manage his own affairs"* is a real fear and it can be met honestly: needing help with a fraud is not the same as being unable to manage. The lever only works while it is unnamed.

18.6 Where the analogy stops

The circuit describes the *mechanism* of the manipulation. It says nothing about the caller, who is a person with reasons, and nothing about the man, whose sixty years are not a defect to be corrected. And there is a boundary worth stating plainly: this chapter describes a structure so it can be recognised. It is not a script, and the parts it names are exactly the parts a competent fraudster already knows.

18.7 What to take from it

- The library is neutral. The same components describe a teacher and a thief, and recognising the shape is the whole defence.
- Attacks are aimed at what you are unwilling to be, not at what you believe.
- A boundary whose trigger has been renamed will never fire. Attach it to what you can see.
- Isolation and hurry are load-bearing parts of the attack, not incidental. Remove either and it weakens.

- The notation names the substrate; the drawing names the wire to cut. Neither alone is enough.

Chapter 19

What Else This Pair Illustrates

Chapter 19.

Nine situations have been worked through in full. The library is not limited to those nine, and this chapter is a map rather than an argument: for each domain, the parts that do the work and the thing the pair lets you say that ordinary language struggles to. None of these is a claim of clinical or scientific validity. They are the domains where these components have earned their keep in this book, and where a reader can go next.

19.1 Love and attachment

Parts: capacitor, current mirror, LC tank, reference, regulator.

Attachment is retention with a long time constant, and much of what people find mysterious about it becomes ordinary once it is described that way — why a person can be gone for a decade and still shape an evening; why a familiar tone lands before any thought; why two people who love each other can still resonate on one frequency and nowhere else. The reference node is the part that explains why leaving a long relationship is disorienting in a way that has nothing to do with missing anybody: the zero point moved.

19.2 Fear, anxiety, and threat

Parts: Schmitt trigger, oscillator, zener, fuse, rail.

A threat system is a comparator with a threshold, and almost every intervention that works is a change to the threshold, the deadband, or the supply — not to the fear. The `|E` register is the fastest one and therefore the first place to work. Naming the trigger condition correctly is what lets the protective rule fire at all.

19.3 Anger and conflict

Parts: resistor, differential pair, LC tank, relay, circuit breaker.

Anger is usually a signal doing a job, and the useful question is what it is a difference *between*. Conflict that repeats with the same shape is a resonance, and resonance is changed by changing `L` or `C`, never by resolving to do better. The resistor is the whole of the "count to ten" advice, and it is more precise than the advice: pressure unchanged, rate constrained, cost paid.

19.4 Habit, discipline, and change

Parts: inductor, oscillator, crystal, PLL, DAC.

The single most useful fact here is that inductance opposes a change in *rate*. Nearly all failed self-improvement asks for the largest possible rate of change and therefore meets the largest possible opposition. The crystal explains why a fixed daily practice holds better than an enthusiastic irregular one, and the PLL explains why you cannot entrain yourself or anyone else by jumping straight to a distant rhythm.

19.5 Learning and skill

Parts: PLL, crystal, ADC, DAC, active load.

Capture range and lock range are the two numbers that make deliberate practice deliberate. Repetition outside the capture range is not practice. Naming a continuous bodily difference in words is what converts practice into something you can steer.

19.6 Grief and loss

Parts: capacitor, leakage resistor, reference, transformer.

Retention persists, leaks, and can be retuned. The reference moved, and the reference is the part that takes longest to move back. The distinction between discharging a feeling and muffling it is the difference between two completely different outcomes from the same behaviour.

19.7 Purpose, meaning, and spiritual life

Parts: rail, regulator, reference, solid arrow, crystal.

This is the `|T` register, and in this framework it is not decoration. Purpose is a *supply*. A board with an unsupplied `|T` rail runs every other stage at reduced margin — which is the mechanism behind a great many things that get called depression, burnout, or a crisis of faith. A practice that keeps you oriented is a reference; integrity is a reference other people set their clocks by; a vow is a relay. Religious and contemplative traditions have been describing these structures for a very long time in their own vocabulary, and the components here are one attempt at a vocabulary that does not require you to accept any of them.

19.8 Burnout, exhaustion, and collapse

Parts: rails, brownout detector, thermal cutoff, sequencer, regulator dropout.

Supply faults are invisible from the output until they are not. The warning tier is an instruction. Restore in sequence — body, connection, plan, output — or the board browns out again.

19.9 Manipulation, propaganda, and coercive control

Parts: current mirror, low-pass, relay, reference, mislabelled fuse.

Covered in Chapter 18. The general form is worth remembering: an attack on a person's reasoning is much harder than an attack on their *classification* and their *isolation*. Rename the situation and the rules attached to the old name will not fire. Keep them alone and hurried and the loop closes. And the reference is the prize — move a person's zero and everything they do afterwards is measured against a world the operator supplied.

19.10 Institutions and groups

Parts: rails, sequencer, reference, ground, crystal, XOR, majority logic.

Everything in this book scales. A team has supply rails (what work can actually start), a reference (what counts as a normal week), a crystal (the person whose steadiness others lock to), and failure modes that look exactly like an individual's: brownout, resonance, a threshold nobody can cross, a rule whose trigger condition was quietly renamed. An organisation's fuse often cannot fire for the same reason a person's cannot — the threshold is above every level the system can reach.

19.11 Teaching and being taught

Parts: transformer, ADC, DAC, PLL, probe.

A transformer changes level without changing content and isolates the two sides — which is the precise statement of why explaining something to a beginner is not a diluted version of the real thing. The ADC and the DAC are the two halves of learning anything: naming the continuous difference, then rendering the name back into practice.

19.12 Self-knowledge

Parts: probe, net label, meter, reference, differential pair.

The probe observes without joining in; the net label turns a bundle of observations into one named wire; the meter insists that a number carry its method; the differential pair removes everything you share with the situation and shows you what is left, which is you; and the reference asks the question that does more work than any other single question in this book — *what am I measuring this against, and who set that zero?*

And the question underneath all of them, from Chapter 2: **whose board are you running?** Which parts of your behaviour did you place, deliberately, and which were installed by somebody else for a purpose that was never yours? The inherited ones are not faults. They are components somebody else chose, still running, and the only way to tell the difference is to read the schematic and ask who designed each stage.

19.13 Theatre, fiction, and ritual

Parts: transformer, oscillator, crystal, PLL, reference, relay.

A play is a deliberately built circuit that runs on an audience. A ritual is a fixed sequence with a crystal at its centre — the same words, the same hour, the same order — and the point of the fixity is that it holds when the participants cannot. Fiction is a transformer: it carries content across a gulf between one person's life and another's

without either side touching the other directly, which is precisely why it can say things that a direct statement cannot.

19.14 The map is not the territory

Two warnings, both of which the framework itself insists on.

First: no part of this is a diagnosis. These are structural descriptions. A structure that fits is not a condition you have, and fitting several of them is not a syndrome.

Second: the components are neutral and the description is not a cure. Knowing that a repeated argument is a resonance does not lower its Q. Knowing that the fuse is mislabelled does not relabel it. The value is that the right lever becomes visible — and then somebody still has to pull it, usually slowly, usually with help.

Third: a map is not a design. Everything in this chapter is description, and description is what the notation does. Turning any of it into a design means placing components on purpose — choosing a threshold, stating a default, deciding what fails safe — and taking responsibility for the fact that you chose them. That is the difference Chapter 2 draws between reporting a state and committing to one, and it is the difference between understanding your life and authoring part of it.

Fourth, and the most hopeful thing in the book: the same expression can be run on radically different boards. Two people with an identical description of what happens to them — same Noetic, same Domain, same Drive, same Register — can live completely different versions of it, because gain, thresholds, clamps, return paths, time constants and reachable boundaries are all decisions, and none of them is in the expression. You cannot change what kind of event you are dealing with. You can change its magnitude, its speed, its reversibility, and whether it fails safe. Most of what determines how a life goes was never in the event.

Part IV

The Schematics

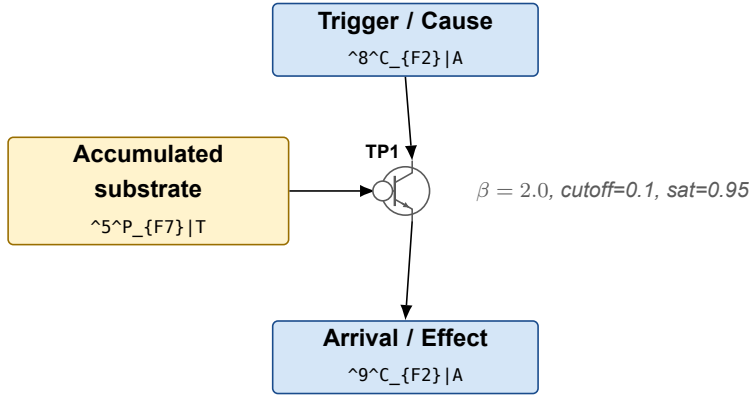
The same circuits, drawn at reference size

Chapter 20

Single-Stage Cognitive Schematics

Circuits 1–5.

20.1 Circuit 1 — The CJT Atom



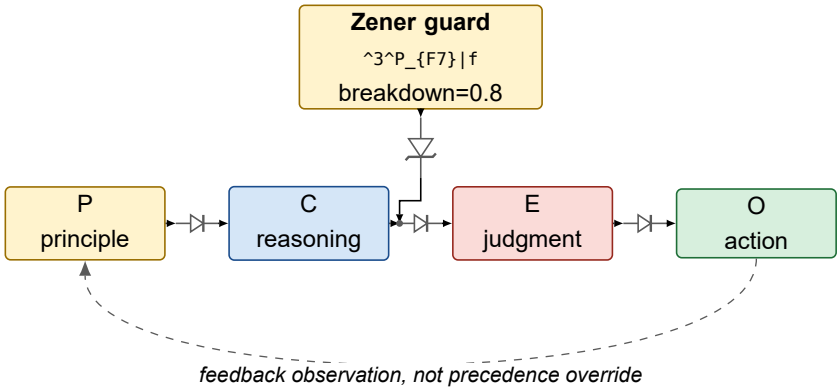
Cognitive roles: trigger enters, accumulated composition biases, a different Idea arrives. Hardware terminals remain a BJT-style drawing; the role labels are the CAS abstraction.

Figure 20.1: The CJT atom: an Above/Cause trigger, shaped by a Female substrate, lands as a Below/Effect. This is a trigger-to-arrival transition, not a synonym for reasoning.

Example. A question reminds a system of a learned procedure. The question is the `^8` trigger. The training and accumulated examples are the `^5` substrate. The procedure brought to Mind is the `^9` effect.

In a life. Someone says your name in a particular tone and you are eight years old in a kitchen. The tone is the `^8` trigger, the eight-year-old is not a memory you fetched — she is the `^9` that arrived — and the household you grew up in is the `^5` substrate that decided what the tone would land as. Change the substrate and the same tone lands as an ordinary question, which is why two siblings hear the same sentence and one of them flinches.

20.2 Circuit 2 — Directed Domain Descent



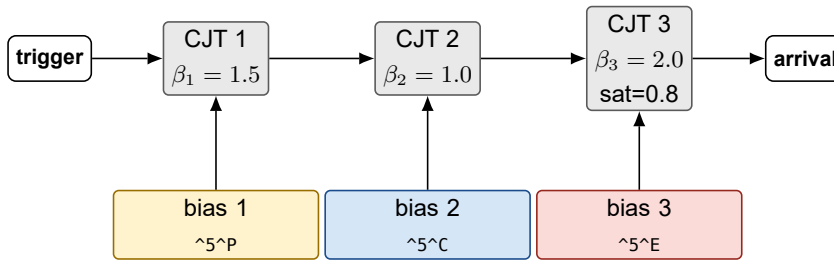
Diodes realize directed admissibility: whether a crossing may occur. A Transformer, when present, controls the level after a crossing is licensed. RPM remains a section-retraction; it is not claimed to be an ideal reverse-blocking diode.

Figure 20.2: Domain Descent uses directed admissibility for P to C to E to O. The O-to-P return is a labeled feedback path, not a casual upward override.

In a life. A conviction you hold in principle reaches your hands as a way of behaving, or it does not. *I believe people deserve a second chance*

is P. *I will hear him out on Thursday* is C. *This is unfair and I am angry about it* is E. *I have written the email* is 0. Each crossing is a diode: it can happen or be refused, and the refusal is not the same event as never having had the conviction.

20.3 Circuit 3 — Gain Cascade with Separate Biases

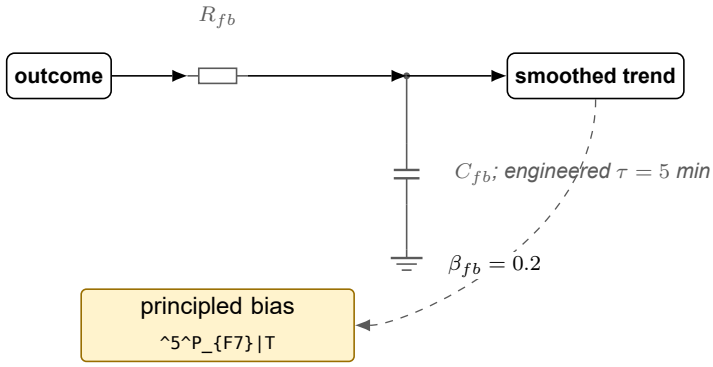


Nominal active-region gain multiplies: $1.5 \times 1.0 \times 2.0 = 3.0$. The final 0.8 saturation ceiling still bounds the output. Each bias is explicit; a diagram must not silently treat one substrate as every stage's local control.

Figure 20.3: A three-stage gain cascade. Product gain is a design calculation; saturation and cutoff remain stage-local limits.

In a life. One small habit, repeated, becomes a disposition. The first honest sentence is effortful. The hundredth is character. This is a cascade, and it explains why single dramatic decisions change so much less than people expect: the gain of one stage is not the gain of the chain, and every stage in the chain has its own ceiling.

20.4 Circuit 4 — RC-Compensated Feedback

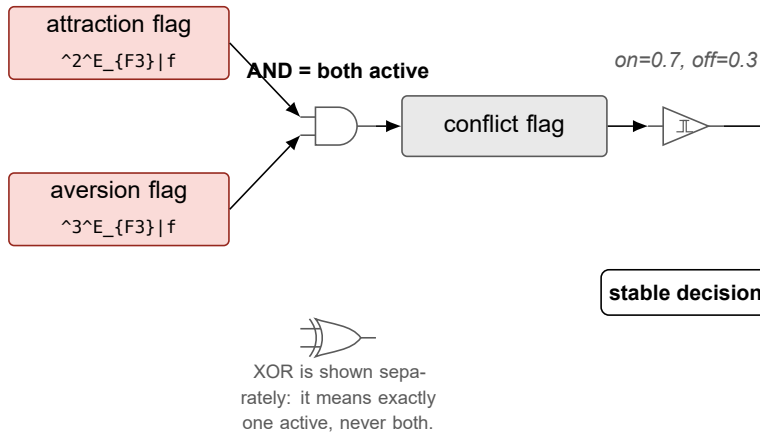


The capacitor integrates recent outcome into a slower ^4 Vibration impression. The dashed path returns only that sampled trend; feedback does not rewrite the output instantaneously.

Figure 20.4: RC compensation smooths an observed outcome before low-gain feedback returns it to the bias path.

In a life. You say something in a meeting, someone's face changes, and by the time you are back at your desk you have revised what you said into something safer. The revision is the feedback resistor: it slows the return of your own output so that it can inform the next thing you say rather than simply echo the last thing. Without it you would repeat yourself all day; with too much of it you never say anything again.

20.5 Circuit 5 — Ambiguity Stabilizer



Continuous activations first cross explicit firing thresholds. AND then detects co-active attraction and aversion. The Schmitt dead-band holds the previous decision while the conflict magnitude wobbles between 0.3 and 0.7.

Figure 20.5: Corrected ambiguity stabilizer: AND detects both-active conflict; Schmitt hysteresis prevents chatter. XOR retains its standard exactly-one truth rule.

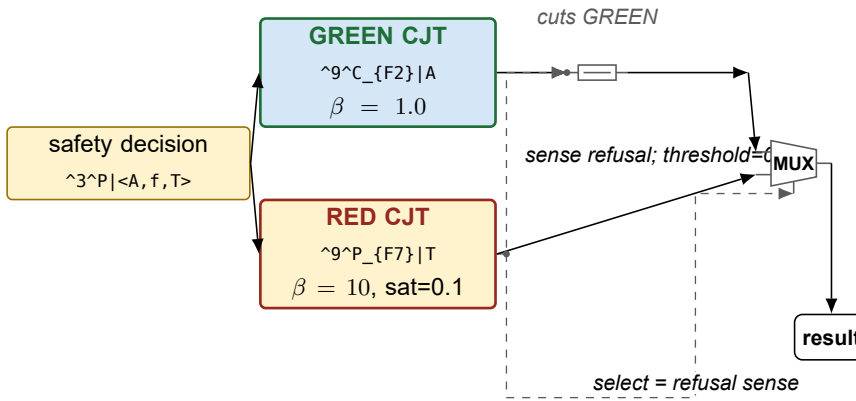
In a life. You want to go and you want to stay, both at once, and you are standing in the hallway unable to move. That is not indecision — it is two flags active at the same time, and the honest reading is *conflict*, not *confusion*. Name both. The hallway is what happens when a person tries to resolve a both-active state by waiting for one of them to disappear.

Chapter 21

Board-Level Cognitive Schematics

Circuits 6–10.

21.1 Circuit 6 — CSB Red Path and Reachable Fuse

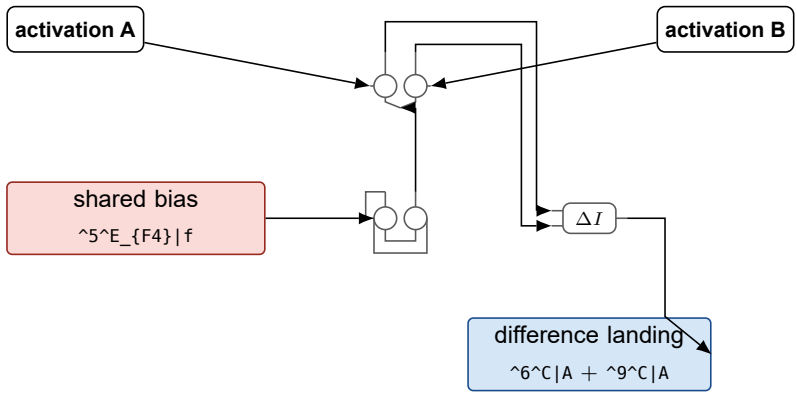


$0.05 < 0.1$: the refusal can reach the fuse threshold before the red stage's clamp. The sense line trips the latch; the fuse severs the green answer path. The refusal path remains available.

Figure 21.1: The corrected CSB red path. A reachable 0.05 fuse threshold sits below the 0.1 refusal clamp and cuts the green path, never the refusal path.

In a life. A small sentence switches something enormous. *I am going to say it tonight* changes a marriage; the sentence itself took a second. The isolation matters: the decision can be made in a calm hour without the consequence reaching back and moving the deciding. Note the direction, always — the large path does not get to steer the small one, however much it feels as though it does.

21.2 Circuit 7 — Differential Comparison

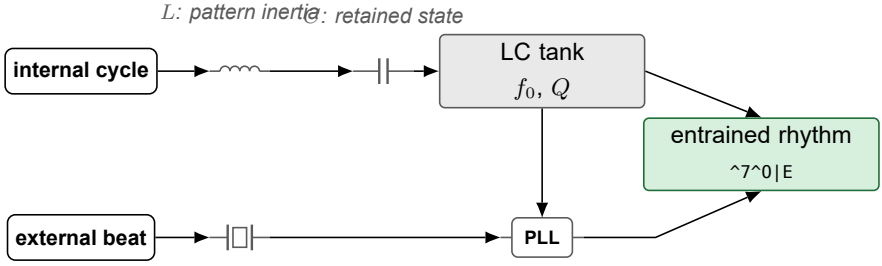


Both inputs must be commensurable activations on one axis. Matched gain and one shared bias reject common-mode content. Decomposition is ^6 Male; the resulting Idea lands as ^9 Below/Effect.

Figure 21.2: Differential comparison rejects what two commensurable inputs share and preserves their difference under one shared bias.

In a life. Two people hear the same criticism. One of them is fine and the other is not, and the difference is not the criticism. Something both of you share — the fact of being criticised, the fact of the week you have both had — is cancelling out, and what is left is the part that is actually about each of you. That is uncomfortable precisely because it is informative.

21.3 Circuit 8 — Rhythm, Resonance, and Entrainment

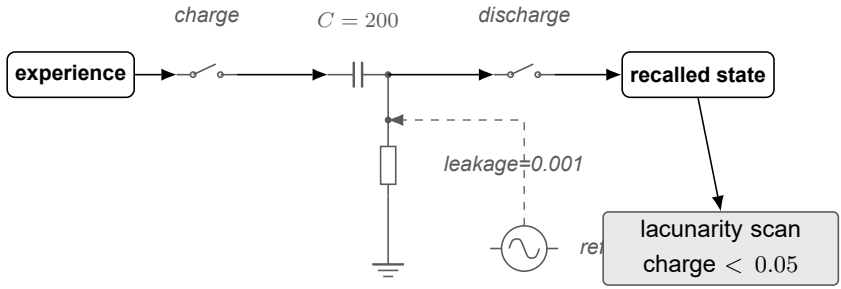


The LC tank expresses ^4 Vibration as resonance: one rate strikes the strongest chord. The PLL expresses ^7 Rhythm as entrainment: repeated external timing conditions the internal cycle. The crystal is a low-drift reference, not a Noetic by itself.

Figure 21.3: The COB separates resonance from entrainment: LC selects the resonant rate; PLL locks the repeating cycle to a reference.

In a life. A practice rhythm and an external rhythm find each other: a new baby's sleep and the household's week; a runner's cadence and a partner's; a team's pace and a new starter's. The lock is gradual and it has a range, which is the practical fact — you cannot drag somebody into a rhythm far from their own, and you can walk them into it in steps.

21.4 Circuit 9 — Retention, Leakage, Refresh, and Lacunarity

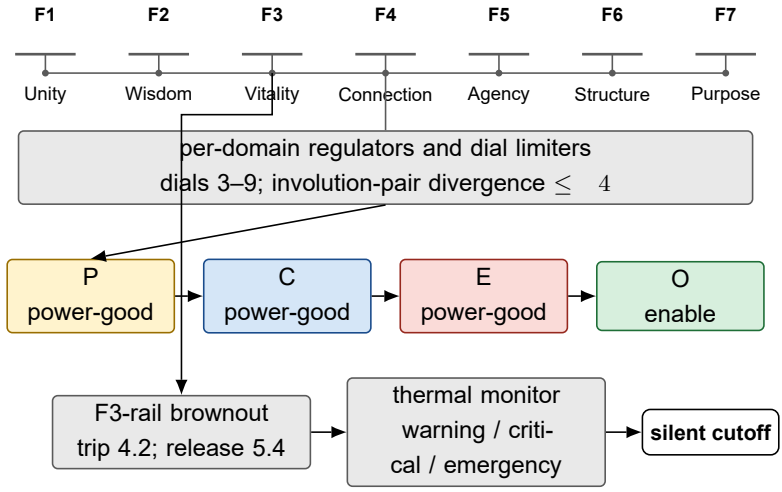


A low reading alone cannot distinguish never-written from leaked-away. Write and refresh history must accompany the charge probe. The capacitor models persistence of composition, not a Mind as a container.

Figure 21.4: A CMB cell retains state, leaks, refreshes, and exposes low-charge addresses; provenance separates true lacunae from forgotten contents.

In a life. You remember, and you are changed by the remembering, and over years the change softens. This is retention with a leak, refreshed by rehearsal, occasionally showing a gap where none was. The gap is the part people fear most, and it is worth saying plainly: a low reading cannot tell you whether something was never written or has faded since. Only the history can.

21.5 Circuit 10 — Drive Supply, Sequence, Brownout, and Silence



Operational output enables only after P, C, and E are power-good. Brownout uses hysteresis. Emergency response cuts all CJTs to silence; it does not improvise an output while the supply is invalid.

Figure 21.5: The CPMB supplies seven Drive rails, regulates them, enforces P-to-C-to-E-to-O startup, derates on brownout, and fails closed to silence at emergency.

In a life. The whole board in one picture. Somebody is running on purpose and structure, has not slept properly in four months, and is still delivering — right up until the afternoon when the lights go out. The output was clean the entire time, which is precisely why nobody looked at the supply.

Part V

Reference

One table to memorise, and the printable deck

Chapter 22

The Complete Memory Deck

One line per part.

Use the first verb as the mnemonic:

Part	Electrical verb	Cognitive verb
Resistor	limit	restrain activation
Capacitor	retain	preserve composition
Inductor	oppose change	preserve an established pattern
Varactor	tune capacitance	tune retention or resonance
LC tank	resonate	favor one repeating rate
Diode	allow one way	enforce directed admissibility
Zener	trip and clamp	enforce a threshold with a ceiling
Transformer	convert level	translate across a licensed Domain seam
Photodiode	transduce light	admit external Idea content through Mind intake
Bridge rectifier	normalize scalar polarity	PED only on a named non-valence scalar
CJT	control and amplify	let bias shape trigger into arrival

Part	Electrical verb	Cognitive verb
Op-amp	compare	amplify a commensurable difference
Differential pair	reject common mode	preserve a matched difference
Current mirror	copy bias	share one comparison condition
Active load	convert difference	land a branch difference as output
Oscillator	repeat	sustain a cycle
Crystal	stabilize frequency	hold a low-drift cadence
PLL	lock	entrain an internal rhythm
AND	require all	detect all-active prerequisites or conflict
OR	require any	accept one or more thresholded flags
NOT	complement	negate one explicit firing flag
NAND	reject all-active	invert an AND result
NOR	require none	accept only when every flag is inactive
XOR	distinguish	accept exactly one active flag
XNOR	agree	detect equal flags, including both-inactive
MUX	choose	route one selected input to one output
DEMUX	dispatch	send one input to one selected output
Relay	isolate and switch	let Cause control a separate Effect path
Schmitt trigger	commit and hold	prevent threshold chatter
ADC	quantize	translate a continuous Register into discrete form
DAC	reconstruct	translate a discrete Register into continuous form

Part	Electrical verb	Cognitive verb
Regulator	restore	maintain a setpoint
Fuse	sever once	permanently stop an unsafe path for the stated lifetime
Circuit breaker	trip and reset	stop a recoverable fault path
Drive rail	supply	make motive potential available
Reference	define zero	set an engineered baseline
Probe	observe	inspect one named point without becoming its signal
Switch	connect	enable or inhibit one path
Meter	measure	report one named quantity by a named method
Net label	name	type the whole state or scalar
Solid arrow	advance	show the main forward path
Dashed arrow	return	show a sampled feedback path

This table covers all 42 frozen component and annotation entries. The printable 56-card deck prints 29 of those entries individually, folds five narrow or derived entries into their parent cards — **Varactor** into Capacitor and LC tank, **Current mirror** into Differential Pair, and **NAND**, **NOR** and **XNOR** into the AND/OR/NOT/XOR family — and groups the eight annotation entries into three study families: Supply & Zero, Observe & Name, and Route & Grammar. Nothing is dropped: the chapter plates and Appendix A of this guide retain all 42 individual entries.

Chapter 23

Printing and Using the Card Deck

Fifty-six cards, seven sheets, one US Letter page per eight cards.

The companion deck is a separate PDF: `CAS_Component_Companion_Cards.pdf`. It carries the whole guide in a form you can hold in one hand at the bench, cut out, shuffle, and lay next to a schematic.

Print it one-sided, at Actual Size, on US Letter. Not "fit to page". The cards are sized so that the body text lands at a readable size on paper, and every sheet carries crop marks — sixteen ticks around the border plus one between each card — so a guillotine or a ruler and a craft knife gives clean edges. Two columns by four rows, eight cards per sheet, 3.61 by 2.34 inches each.

Scale the page and three things break at once: the text drops below the size it was set at, the cut marks stop lining up, and the cards stop being the same size as each other. If your printer insists on fitting the page, turn that setting off rather than accepting it.

What is in the deck

Cards	What they are
1–5	Five guides: the 1:1 rule, the Domain legend, reading a CAS state, operators and firing, and the four drift stops.

Cards	What they are
6–34	Twenty-nine component cards, one per part, with symbol, role, example and guardrail.
35–42	Eight annotation cards: the Drive rail, the reference node, the probe, the switch, the meter, the typed net label, and the two arrow kinds.
43–52	Ten circuit cards, each a compact signal chain for one cognitive schematic.
53–56	Four use cards: how to study the deck, the claim boundary, printing, and building your own circuit.

Five entries are taught on their parent cards rather than printed separately, because they are narrow or derived and a card of their own would teach less than the sentence it replaced: **Varactor** on the capacitor and LC-tank cards, **Current mirror** on the differential-pair card, and **NAND**, **NOR** and **XNOR** on the XOR card. All forty-two entries keep their own plate and their own row in the memory deck — the deck of cards is a study aid, not the reference.

Sorting the deck

Cards carry a number, so keep them in number order the first time through. After that, two other orders earn their keep:

- **By job.** Group the limiters (resistor, capacitor, inductor), the one-way parts (diode, Zener), the amplifiers (CJT, op-amp, differential pair, active load), the timing parts (oscillator, crystal, PLL), the logic parts, and the annotations. Building a circuit from a shuffled deck is a good test of whether you know what each part is *for*.
- **By badge.** Sort into ISO, ENG and PED. The ENG pile is the one worth re-reading: those are the mappings that need a number you have to choose, and choosing it badly is the usual failure.

Studying from the deck

Three passes, in this order, and each one is a different question:

1. **Name the job from the symbol alone.** No text. If you cannot say what the part does before you read the card, the shape has not stuck yet.
2. **Recite the cognitive role and the badge.** Saying "resistor, limits activation flow, ENG" out loud is the whole exercise. The badge is not decoration — it tells you how far the mapping carries.
3. **Explain the example and the guardrail without looking.** This is the pass that matters. The guardrail is the line that tells you where the analogy stops, and knowing it is the difference between using this library and abusing it.

A memorised metaphor is not enough. The role correspondence and the boundary are the content; the metaphor is only how you reach them.

Appendix A

Design Notes and Frozen Inventory

Appendix A.

Working title: Illustrated Electrical Symbols and Cognitive Schematics

Status: maintenance inventory. Part of this guide, and the place to start if you are extending it.

Scope: one functional-role map per symbol. Component symbols are conventional circuit notation adapted to the book's visual grammar; they are not a new CAS ontology and do not imply one component = one Noetic.

A.1 Mapping labels used on the plates

- **ISO — exact structural isomorphism:** one-to-one at the level of functional roles; the relation has the same form.
- **ENG — engineered analog:** the role correspondence is explicit, but a cognitive quantity, control input, or conservation rule is stipulated by the design.
- **PED — pedagogical analogy:** the resemblance teaches only; no quantitative transfer.

A.2 Symbol plates

S0 — Signal and annotation legend

- Terminal / port node — named source or destination boundary (`PORT`)
- Domain color key — P amber, C blue, E red, O green; labels and shapes remain authoritative in grayscale
- Items 36, 37, and 40–42 below are demonstrated here and remain numbered under S6 for deck traceability.
- This guide's printable card deck keeps 56 total cards by folding five narrow or derived entries into their parent cards: Varactor into Capacitor/LC, Current mirror into Differential Pair, and NAND/NOR/XNOR into the AND/OR/NOT/XOR logic family. The chapter plates and this inventory still preserve all 42 individual entries.
- **Deck numbers and inventory numbers agree only from deck card 28 onward.** Cards 6–27 were numbered consecutively across the five folded entries, so deck 9 is inventory 5 (LC tank; Varactor skipped) and deck 18 is inventory 15 (Active load; Current mirror skipped). Never read a deck number below 28 as an inventory index.

S1 — Flow and state retention

1. Resistor — activation-current limiter (`R`)
2. Capacitor — state retention / accumulated composition (`C`)
3. Inductor — pattern inertia (`L`)
4. Varactor — tunable state retention (`D` / `CV`)
5. LC tank — resonant state exchange (`Z` / composite)

S2 — Direction, threshold, and sensing

6. Diode — directed admissibility guard (`D`)
7. Zener diode — threshold conduction and clamp (`ZD`)

8. Transformer — domain impedance matching (`T`)
9. Photodiode — stimulus intake transducer (`PD`)
10. Bridge rectifier — polarity normalization only (`BR`); **PED**, never a `^3` Negative-to-`^2` Positive converter

S3 — Active and comparative parts

11. Cognitive Junction Transistor — controlled trigger-to-arrival transition (`Q1` / `CJT`)
12. Op-amp — high-gain differential stage with feedback (`U1`)
13. Differential pair — matched comparison stage (`Q2`, `Q3`; composite)
14. Current mirror — replicated bias (`Q4`, `Q5`; composite)
15. Active load — current-to-difference conversion (`Q6` / composite)

S4 — Rhythm and register conversion

16. Oscillator — sustained `Phi` cycle (`OSC`)
17. Crystal — low-drift timing reference (`Y`)
18. Phase-locked loop — entrainment to an external reference (`U2` / `PLL`; composite)
19. ADC — register translation with quantization (`U`)
20. DAC — register translation with smoothing (`U`)

S5 — Logic and routing

21. AND
22. OR
23. NOT
24. NAND
25. NOR
26. XOR
27. XNOR / equivalence detector
28. MUX
29. DEMUX

30. Relay
31. Schmitt trigger

Logic correction: standard XOR is true when exactly one input is true. A both-active conflict detector is AND. XNOR reports equal inputs, including both-active and both-inactive; it needs an OR/activity qualifier if used to isolate both-active conflict.

S6 — Regulation, protection, and annotations

32. Voltage regulator
33. Fuse
34. Circuit breaker
35. Drive rail
36. Reference / rest node
37. Probe / test point
38. Switch / enable
39. Meter / CAS-Ohm mode marker
40. Net label (typed CAS state on a wire)
41. Solid forward path
42. Dashed feedback path

A.3 Schematic plates

1. **CJT atom** — `^8` Above/Cause trigger + `^5` Female substrate → `^9` Below/Effect arrival.
2. **Domain descent** — directed admissibility across `P` → `C` → `E` → `0`, plus explicit `0` → `P` feedback path.
3. **Gain cascade** — multi-stage CJTs; stage gain and bias separation.
4. **RC feedback** — sampled output returned through a low-pass compensation path.
5. **Ambiguity stabilizer** — AND conflict detector + Schmitt hysteresis; standard XOR shown separately as exclusivity detector.

6. **CSB red path** — clamped refusal stage, reachable fuse threshold, and output-path polarity.
7. **CDAB differential pair** — commensurable input axes, shared bias, matched gain, difference output.
8. **COB rhythm circuit** — LC tank + PLL + crystal reference.
9. **CMB retention cell** — capacitor, leakage, charge/discharge, refresh, and lacunarity probe.
10. **CPMB supply** — seven Drive rails, regulators, `P` → `C` → `E` → `0` sequencer, brownout hysteresis, and silent emergency cutoff.

A.4 Non-negotiable notation rules

1. Every signal wire carries a typed state or explicitly named scalar/control quantity.
2. Solid arrows are forward signal paths. Dashed arrows are feedback paths.
3. Color reinforces Domain but never carries meaning alone.
4. + means co-presence. × requires a named emergent result. |⟨…⟩ is register-set specification.
5. `^3` Negative is repulsion, not decomposition. `^6` Male is decomposition.
6. `^8` Above/Cause is associative triggering, not reasoning. A reasoned landing uses `^6` and `^9` as appropriate.
7. ADC/DAC examples hold Noetic and Domain fixed when demonstrating pure Register translation.
8. Parameter values on every plate are design choices, not measured cognitive or model results.

A.5 Resolved guardrails retained for maintenance

- The book's CJT labels Cause/Bias/Effect are a cognitive functional abstraction, not the conventional BJT small-signal

input/output terminal convention. Plates must show both labels without claiming transistor terminal identity.

- CAS firing thresholds for Boolean gates are design stipulations; continuous open-simplex activations are not intrinsically binary.
- Standard ground denotes an electrical reference. Calling it a cognitive `^0` rest state is an engineered convention, not a canonical identity; use **Reference / rest node** on the plate.
- The bridge rectifier entry remains pedagogical unless a non-valence scalar polarity is defined. Do not map it as aversion-to-attraction conversion.
- ADC and DAC are separately directed Register translations. Quantization makes ADC lossy, so the DAC is not called its inverse.
- Composite parts with no stable one-line CCDL constructor use a labeled composite schematic or documented call-shaped form rather than invented syntax. Preserve the book's defined `Differential<...>`, `mirror ... replicate(...)`, `active_load ... decompose(...)`, and `LC<...>` forms.

A.6 Card deck accounting

- Cards 1–5: five reading and claim-boundary guides, including the Domain/color/label legend.
- Cards 6–34: 29 individually printed component cards; the five folded roles above are cross-referenced on their parent cards.
- Cards 35–42: eight annotation cards, taught as three families: **Supply & Zero** (#35–36), **Observe & Name** (#37 and #39), and **Route & Grammar** (#38 and #40–42).
- Cards 43–52: ten compact cognitive-state circuit cards.
- Cards 53–56: four study, claim, print, and construction cards.
- Total: **56 cards = 7 US-Letter sheets** $\times$ **8 cards**.

A.7 Automated checks over this guide

Run from `build/`. All of them must pass before this guide ships.

Script	What it proves	What it does not prove
`audit_plate_connectivity.py`	Reviewed source strings — terminal coordinates, typed states, Domain styles, symbol conventions — are still present.	Anything nobody wrote down. It is a regression lock, not a verifier.
`audit_plate_geometry.py`	Reads the **compiled** PDF of each schematic and fails on any wire endpoint that lands on no other ink (tolerance 1.25 mm). Intentional free ends are listed with a reviewed reason in `dangling_whitelist.json`; a whitelist entry that becomes connected also fails, so the list cannot rot.	The seven symbol plates, which draw each glyph in isolation by design.
`audit_symbol_invariants.py`	Each glyph is electrically itself: bridge diodes point at the node they feed, the current mirror is diode-connected and shares its bias, the Schmitt hysteresis mark stays inside the triangle, the switch is drawn open, and the relay linkage, breaker trip square and Zener bar are attached.	Aesthetics, or conformance to a cited standards document.

Script	What it proves	What it does not prove
`test_plate_negative.py`	All three audits above can actually fail: it breaks the library once per historical defect and requires the matching audit to report it.	Nothing further; it exists so the other three cannot pass vacuously.

`verify_plates.py` compiles all 17 figures standalone; `verify_cards.py` checks the 56-card deck (numbering, one badge per card, page count, US-Letter size, and every semantic guard); `test_badge_guard.py` proves the compile-time badge check rejects a hybrid label; and `verify_deliverables.py` cross-checks the finished guide against the finished deck.

A.8 Open review items

- **DAC trapezoid convention.** The wide parallel side now marks the multi-bit port on both converters, with short parallel stubs on the digital side and the `A/D` / `D/A` text naming the direction. This follows the common databook convention; confirm it against the house standard before changing it again.
- **Bridge rectifier node convention.** Left and right corners are AC, top is DC+ and bottom is DC-. The four diode orientations are now *derived* from that convention by `audit_symbol_invariants.py` rather than frozen as literals.
- **Cover spine for the current page count.** Regenerate the cover wrap whenever the interior page count changes; the paperback spine factor is 0.002252 in per page.